

RapidPD

Rapid Human and Pet Presence Detection System for Smart Vehicles via Wi-Fi

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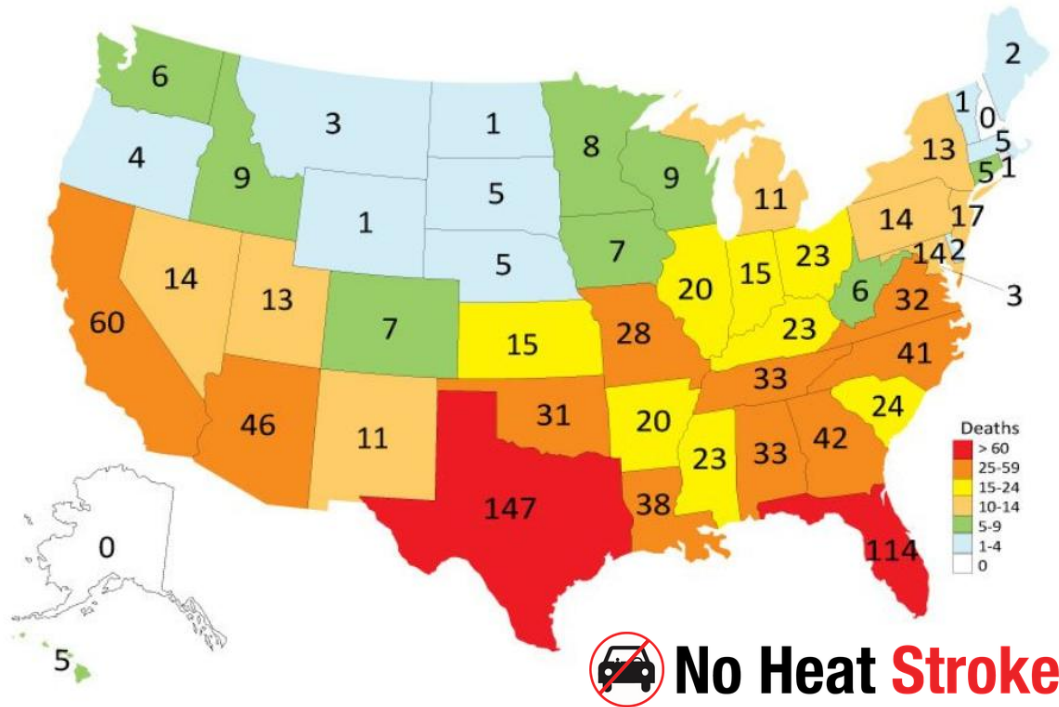
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Widely Concerned In-Vehicle Safety Issue



Pediatric Vehicular Deaths by State



more than 900 children

heatstroke in parked cars

 **No Heat Stroke**

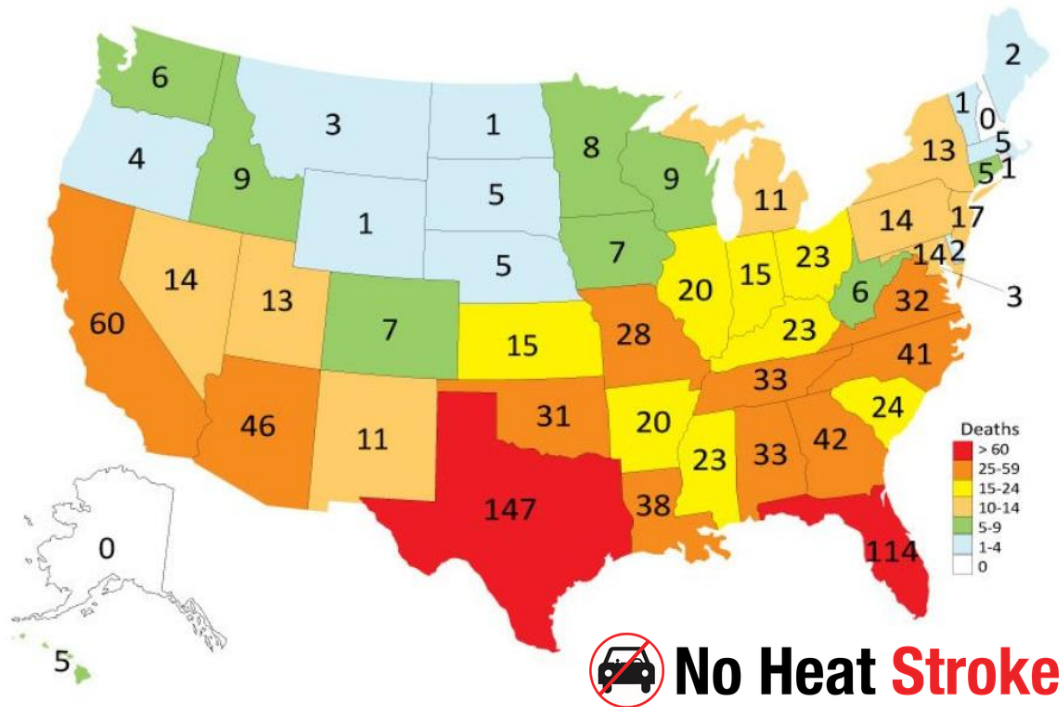
[1] J. Nall. (2024, Sep.) No heat stroke. [Online]. Available: <https://www.noheatstroke.org>

[2] 2024 Euro NCAP. (2023, Dec.) Test and assessment protocol – child presence detection. [Online]. Available: <https://www.euroncap.com/media/79888/euro-ncap-cpd-test-and-assessment-protocol-v12.pdf>

Widely Concerned In-Vehicle Safety Issue



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Effective in-vehicle
presence detection methods
are required

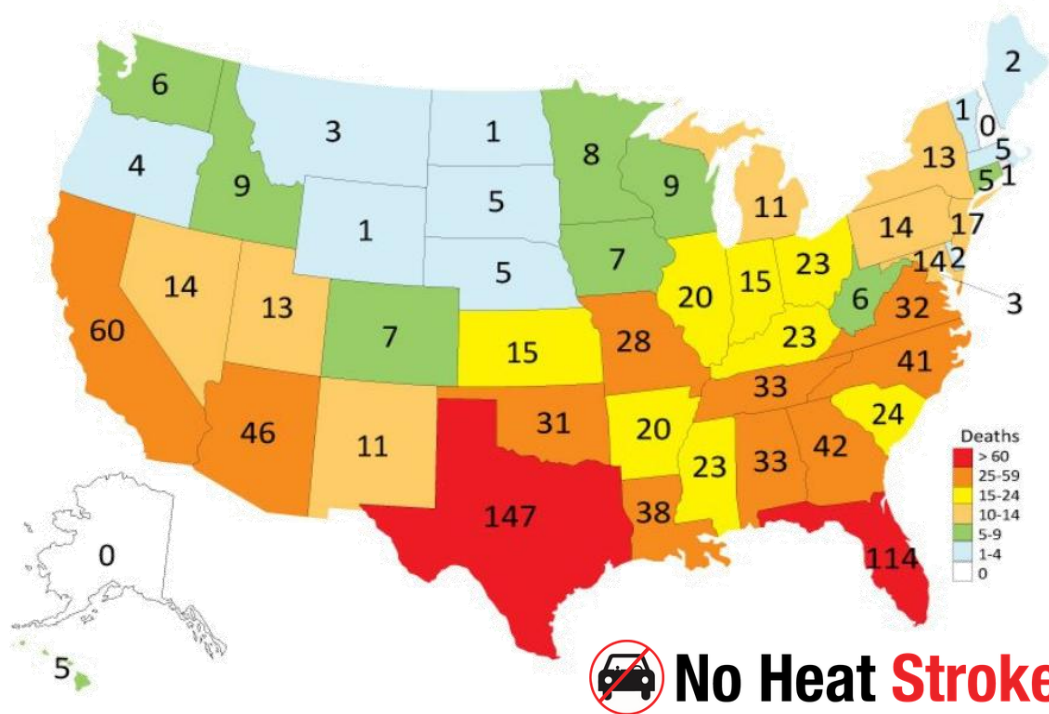
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Widely Concerned In-Vehicle Safety Issue



Pediatric Vehicular Deaths by State



**TEST AND ASSESSMENT PROTOCOL –
CHILD PRESENCE DETECTION**

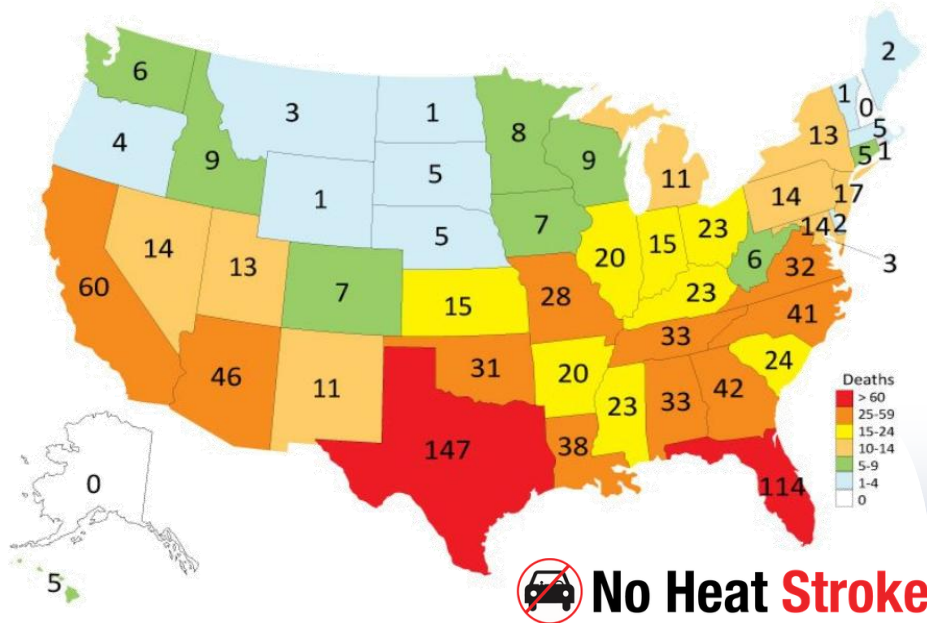
[1] J. Nall. (2024, Sep.) No heat stroke. [Online]. Available: <https://www.noheatstroke.org>

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Widely Concerned In-Vehicle Safety Issue



Pediatric Vehicular Deaths by State



 **No Heat Stroke**

FOR SAFER CARS
EURO NCAP



**TEST AND ASSESSMENT PROTOCOL –
CHILD PRESENCE DETECTION**

Practical needs, together with regulatory mandates,
are driving new vehicles to be equipped with Presence Detection systems.

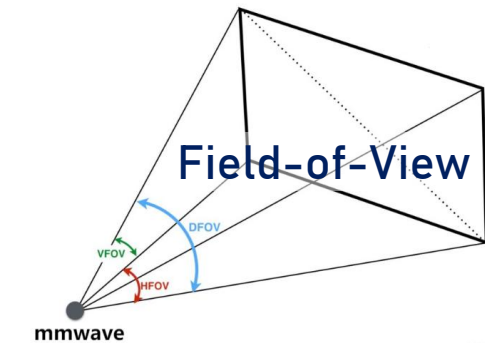
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Challenges

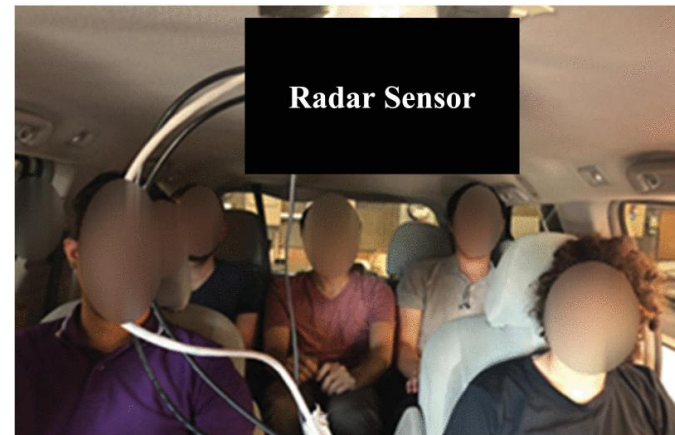


Limited Coverage



Poor Lighting Conditions

High System Cost

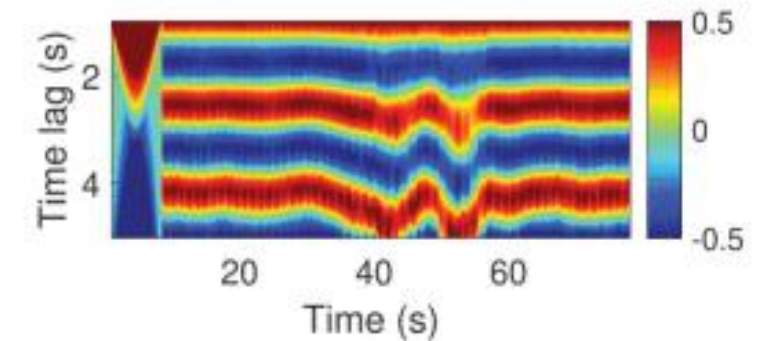
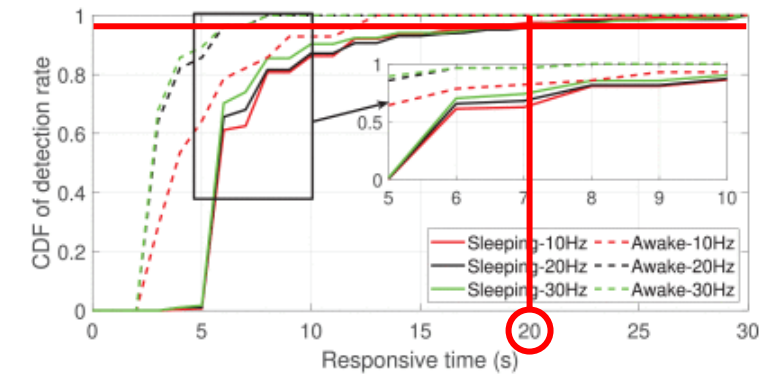


IMAGEVK-74 | 4D Millimeter Wave Imaging Kit

\$3582.02



Slow Response time



- pre-training requirements, privacy concerns, etc.

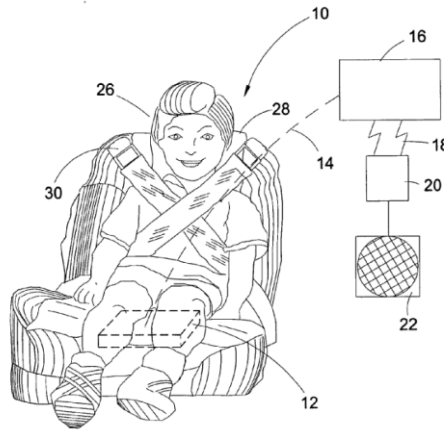
[1] J. Jaworek-Korjakowska, A. Kostuch and P. Skrch, "SafeSO: Interpretable and Explainable Deep Learning Approach for Seat Occupancy Classification in Vehicle Interior," 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), Nashville, TN, USA, 2021, pp. 103-112

[2] H. Abedi, M. Ma, J. He, J. Yu, A. Ansariyan and G. Shaker, "Deep Learning-Based In-Cabin Monitoring and Vehicle Safety System Using a 4-D Imaging Radar Sensor," in IEEE Sensors Journal, vol. 23, no. 11, pp. 11296-11307, 1 June, 2023

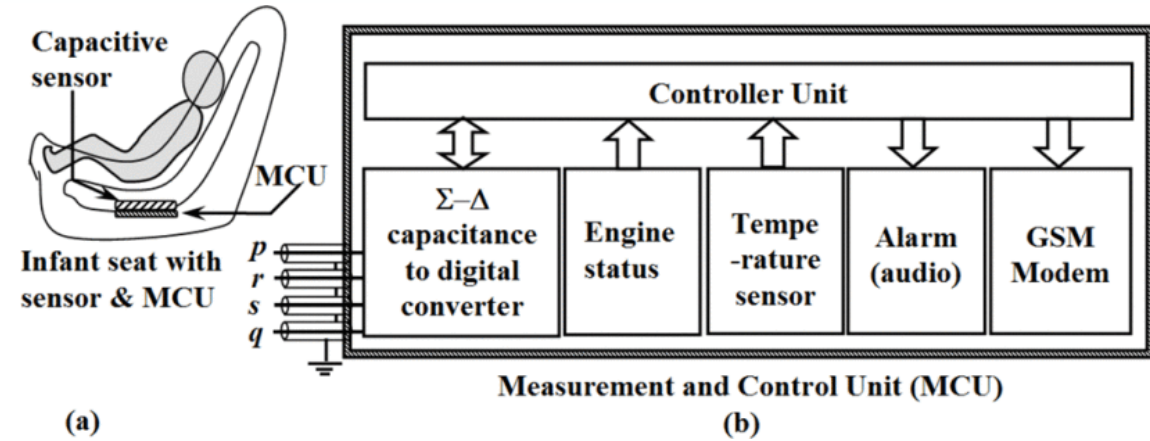
[3] IMAGEVK-74 | 4D Millimeter Wave Imaging Kit, Oct. 2022, [online] Available: https://www.minicircuits.com/WebStore/imagevk_74.html.

[4] Zeng, Xiaolu and Wang, Beibei and Wu, Chenshu and Regani, Sai Deepika and Liu, KJ Ray, "WiCPD: Wireless child presence detection system for smart cars," IEEE Internet Things J., vol. 9, no. 24, pp. 24 866-24 881, 2022

Child Presence Detection Solutions



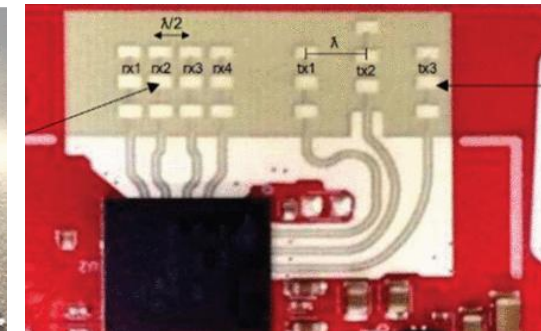
Pressure Sensor
Confused by other objects



Capacitive Sensor
Detection range limited



Embedded Vision
Lighting conditions limited



(mmWave) Radar
High equipment costs

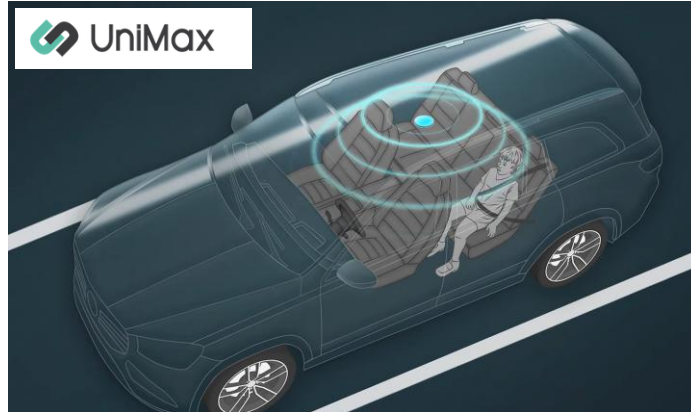
[1] L. Davis, "Child carseat alert system," Jul. 31 2007, uS Patent 7,250,869.

[2] A. Ranjan and B. George, "A child-left-behind warning system based on capacitive sensing principle," 2013 IEEE International Instrumentation and Measurement Technology Conference (I2MTC), Minneapolis, MN, USA, 2013, pp. 702-706

[3] H. Cai, D. Lee, H. Joonkoo, Y. Fang, S. Li and H. Liu, "Embedded vision based automotive interior intrusion detection system," 2017 IEEE International Conference on Systems, Man, and Cybernetics (SMC), Banff, AB, Canada, 2017, pp. 2909-2914

[4] H. Abedi, S. Luo, V. Mazumdar, M. M. Y. R. Riad and G. Shaker, "AI-Powered In-Vehicle Passenger Monitoring Using Low-Cost mm-Wave Radar," in IEEE Access, vol. 10, pp. 18998-19012, 2022

Wi-Fi Based Solutions



UniMax , 2021
Commercial CPD



(a)



⑧ Location of Baby # Location of TX # Location of RX

WiCPD, 2022
High Detection Rate

A promising solution
for CPD:

- Extreme accuracy
- Complete coverage without blind-spots
- Low cost
- Privacy-friendly

[1] UniMax Electronics Inc. (2024, May) Wi-fi child presence detection. [Online]. Available: https://www.unimax.com.tw/en/product_detail/76

[2] X. Zeng, B. Wang, C. Wu, S. D. Regani and K. J. R. Liu, "WiCPD: Wireless Child Presence Detection System for Smart Cars," in IEEE Internet of Things Journal, vol. 9, no. 24, pp. 24866–24881, 15 Dec.15, 2022

Latency Requirement of CPD Systems



Euro NCAP

The time between locking and the initial warning must be as short as possible. However, a delay of no more than 10 seconds is permissible to allow the system to assess occupancy.

UniMax, 2021

The Wi-Fi recognition technology can identify whether there is biological movement in the car within one minute, even detect chest breathing movement of 1~6mm. Because of this technology, Wi-Fi CPD can accurately identify a baby's movement in the car.

WiCPD, 2022

children aging from 4 to 50 months. The results show that WiCPD can achieve 100% detection rate within 8 s when the child is awake/in-motion and 96.56% detection rate within 20 s for a static/sleeping child. Extensive experiments also

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A Critical Issue: Reducing Detection Latency

Latency Requirement of CPD Systems



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UniMax, 2021

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RapidPD, 2025

demonstrate that RapidPD achieves a detection accuracy of 99.05% and a true positive rate of 99.32% within a 1-second time window at a low sampling rate of 20 Hz. These findings represent a significant

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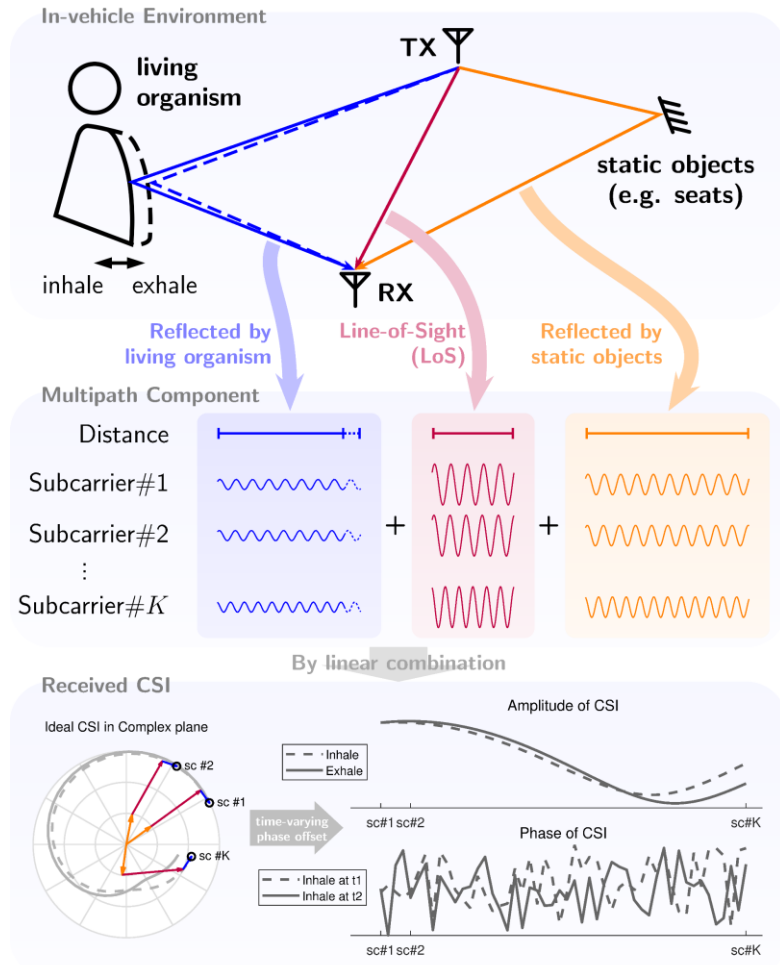


a CPD solution with practical real-world applicability

Main Contributions of RapidPD



Model for Time-varying Envs in Subcarriers

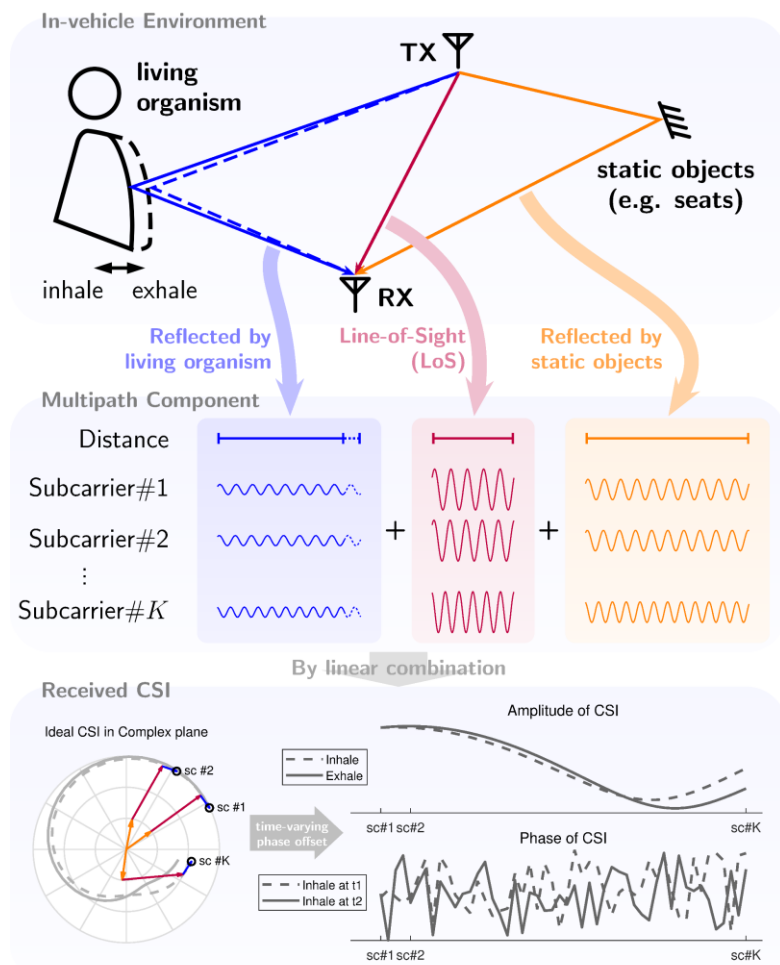


- A CSI model describes time-varying environments
- Theoretical guidance is provided for utilizing the information in subcarrier dimension.

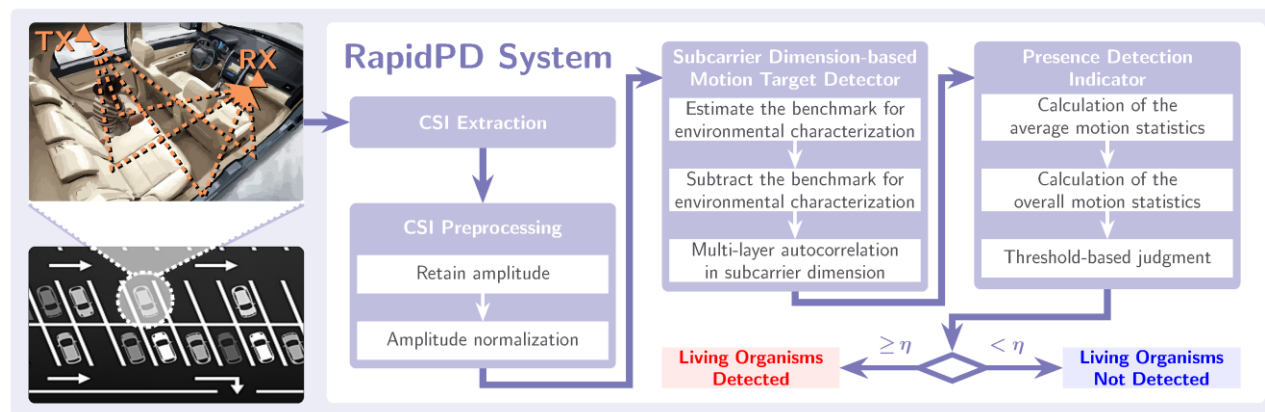
Main Contributions of RapidPD



Model for Time-varying Envs in Subcarriers



A Complete Presence Detection Pipeline with Ultra-Short Time Windows

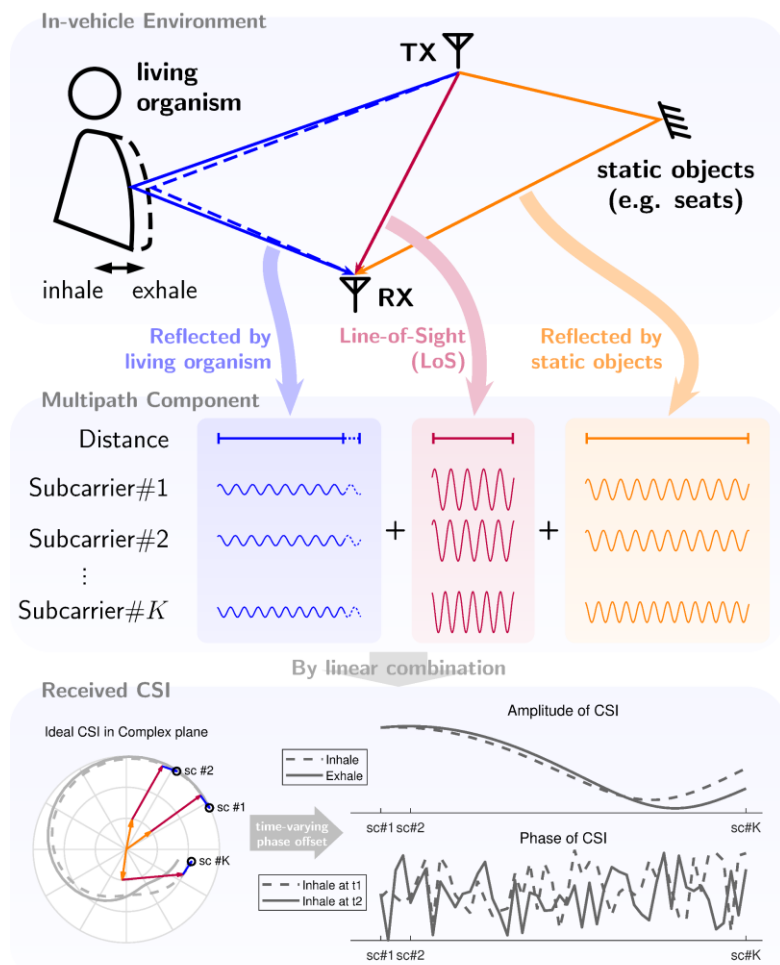


- A complete presence detection pipeline
- Accurate detection within a 1-second time window.

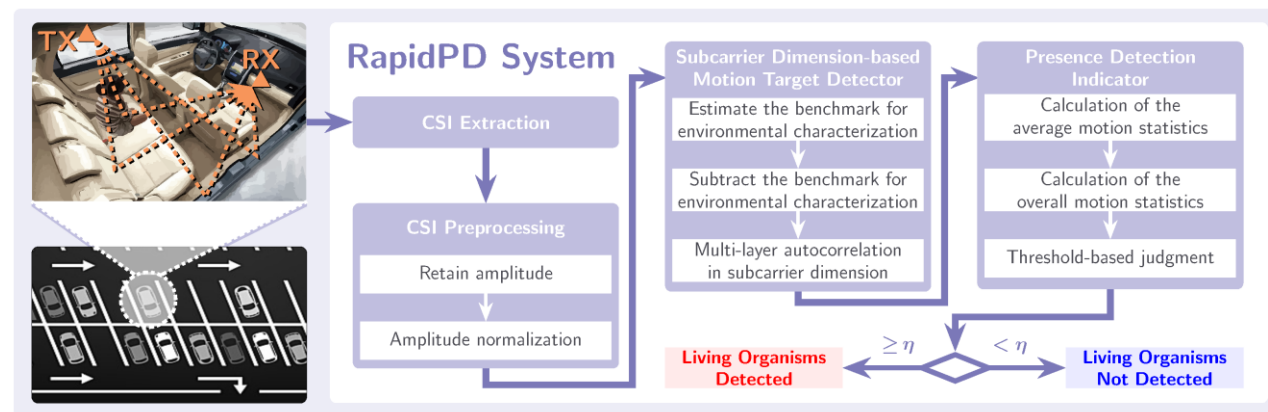
Main Contributions of RapidPD



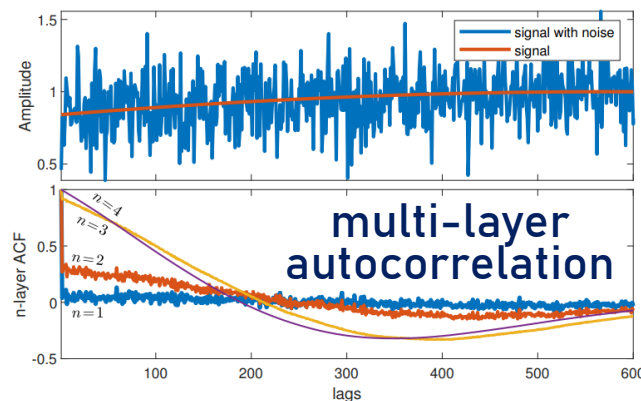
Model for Time-varying Envs in Subcarriers



A Complete Presence Detection Pipeline with Ultra-Short Time Windows



Weak Signal Enhancement

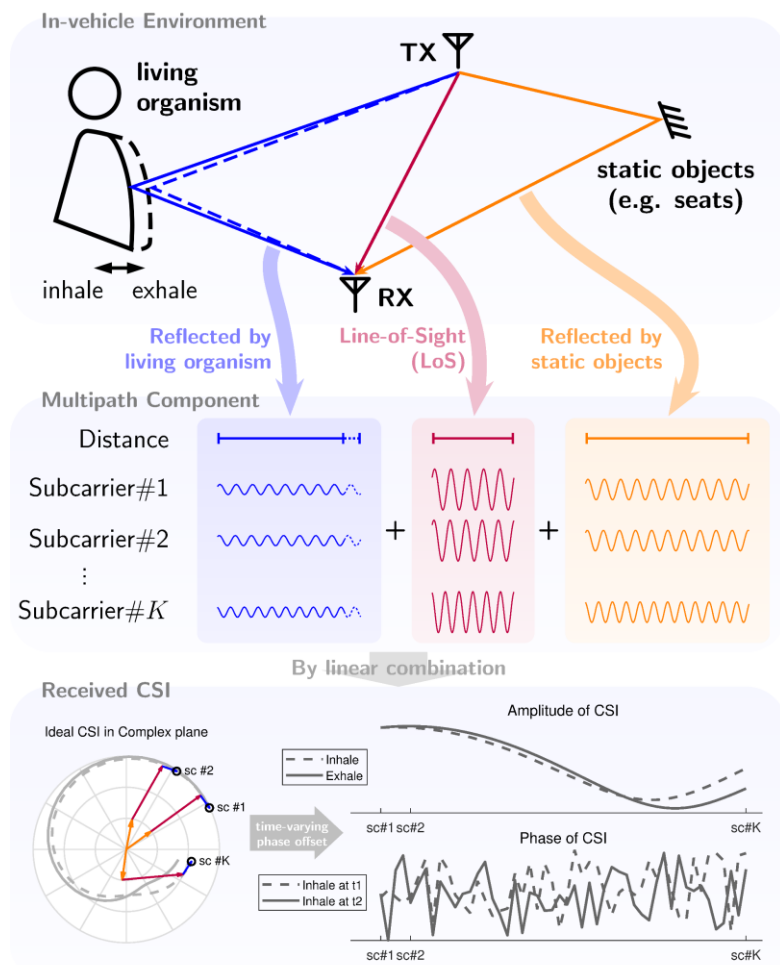


- multi-layer autocorrelations is applied to detect weak signals in concealed locations

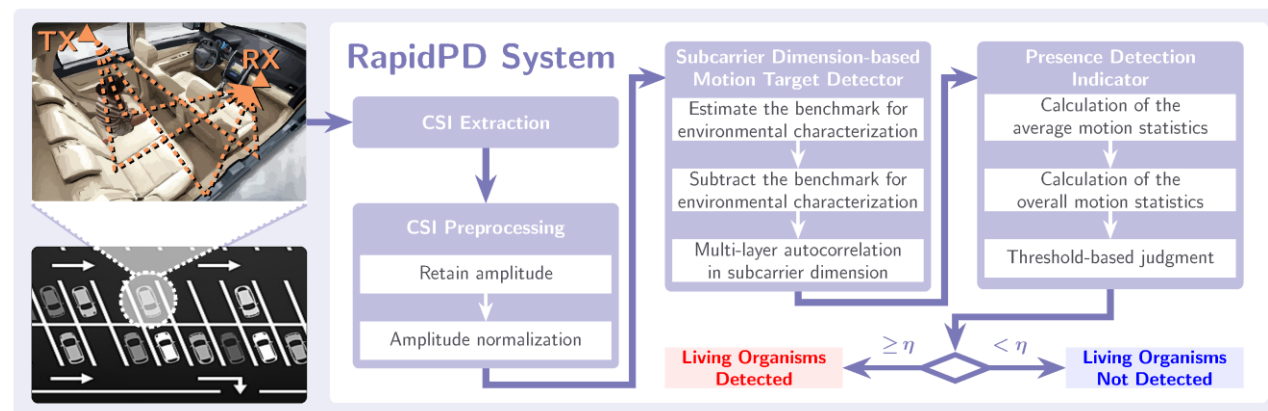
Main Contributions of RapidPD



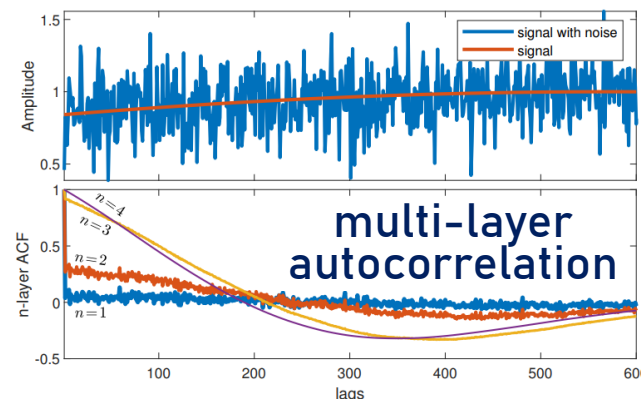
Model for Time-varying Envs in Subcarriers



A Complete Presence Detection Pipeline with Ultra-Short Time Windows



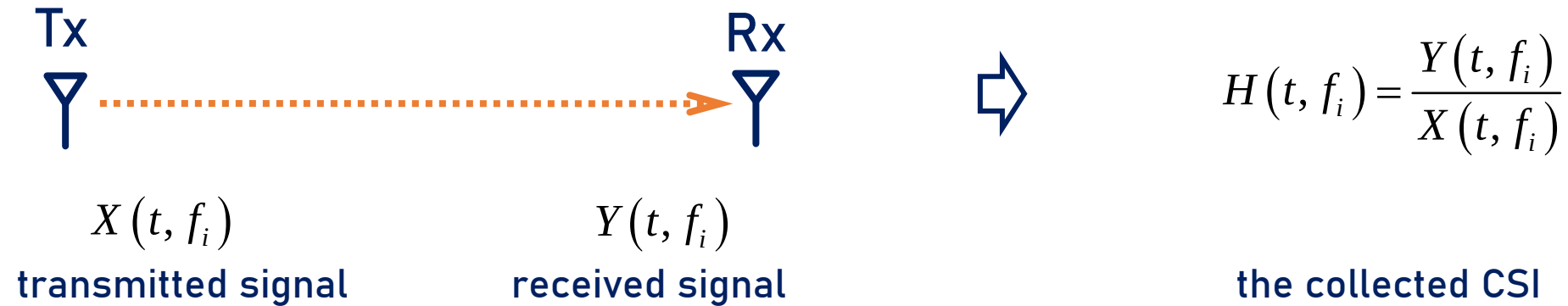
Weak Signal Enhancement



Outstanding Performance

RapidPD (with multi-layer ACF)	
TPR	FPR
99.32%	1.64%
extensive experimental results	

Model for Time-varying Envs in Subcarriers

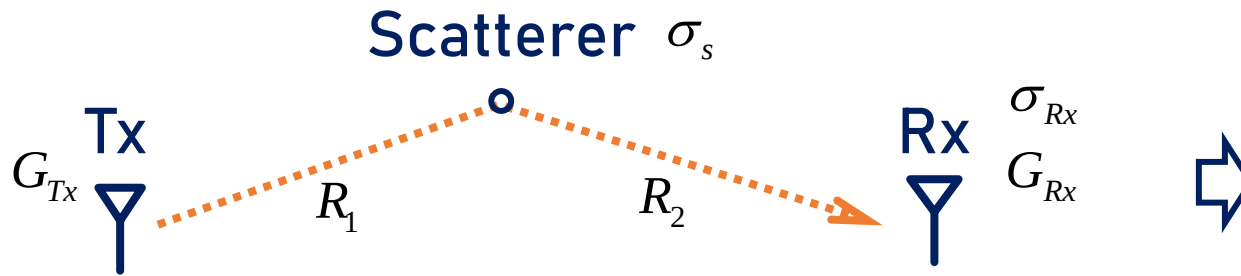


Model for Time-varying Envs in Subcarriers



$$H(t, f_i) = \frac{Y(t, f_i)}{X(t, f_i)}$$

A propagation path with a **single** scatterer:



$$|Y| = G_{Tx} G_{Rx} \frac{\sigma_s}{4\pi R_1^2} \frac{\sigma_{Rx}}{4\pi R_2^2} |X| \quad \text{Power}$$

$$\angle Y = \frac{2\pi f_i}{c} (R_1 + R_2) + \pi + \angle X \quad \text{Phase}$$

radar cross-section σ_s

effective aperture σ_{Rx}

length of 1st path segment R_1

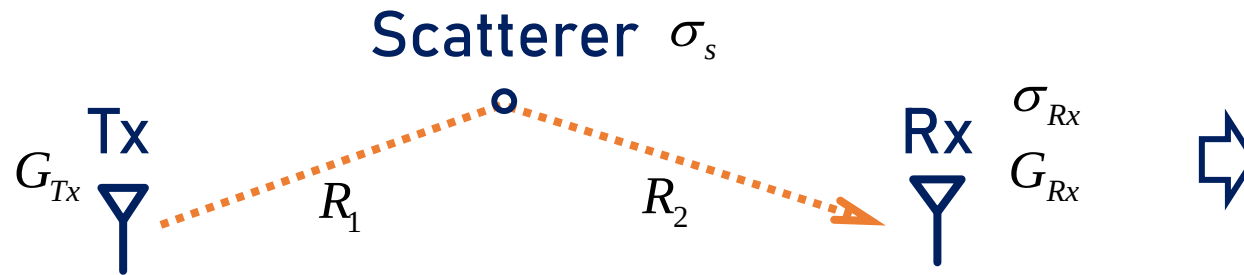
gain of Antenna G

length of 2nd path segment R_2

Model for Time-varying Envs in Subcarriers



$$H(t, f_i) = \frac{Y(t, f_i)}{X(t, f_i)}$$



$$|Y| = G_{Tx} G_{Rx} \frac{\sigma_s}{4\pi R_1^2} \frac{\sigma_{Rx}}{4\pi R_2^2} |X|$$

$$\angle Y = \frac{2\pi f_i}{c} (R_1 + R_2) + \pi + \angle X$$

A propagation path with **multiple** scatterers:



the contribution of path l to CSI

$$H_l = G_{Tx} G_{Rx} \prod_{m=1}^M \frac{\sigma_m}{4\pi R_m^2} \exp \left[j \left(\frac{2\pi f_i}{c} \sum_{m=1}^M R_m + (M-1)\pi \right) \right]$$

Model for Time-varying Envs in Subcarriers

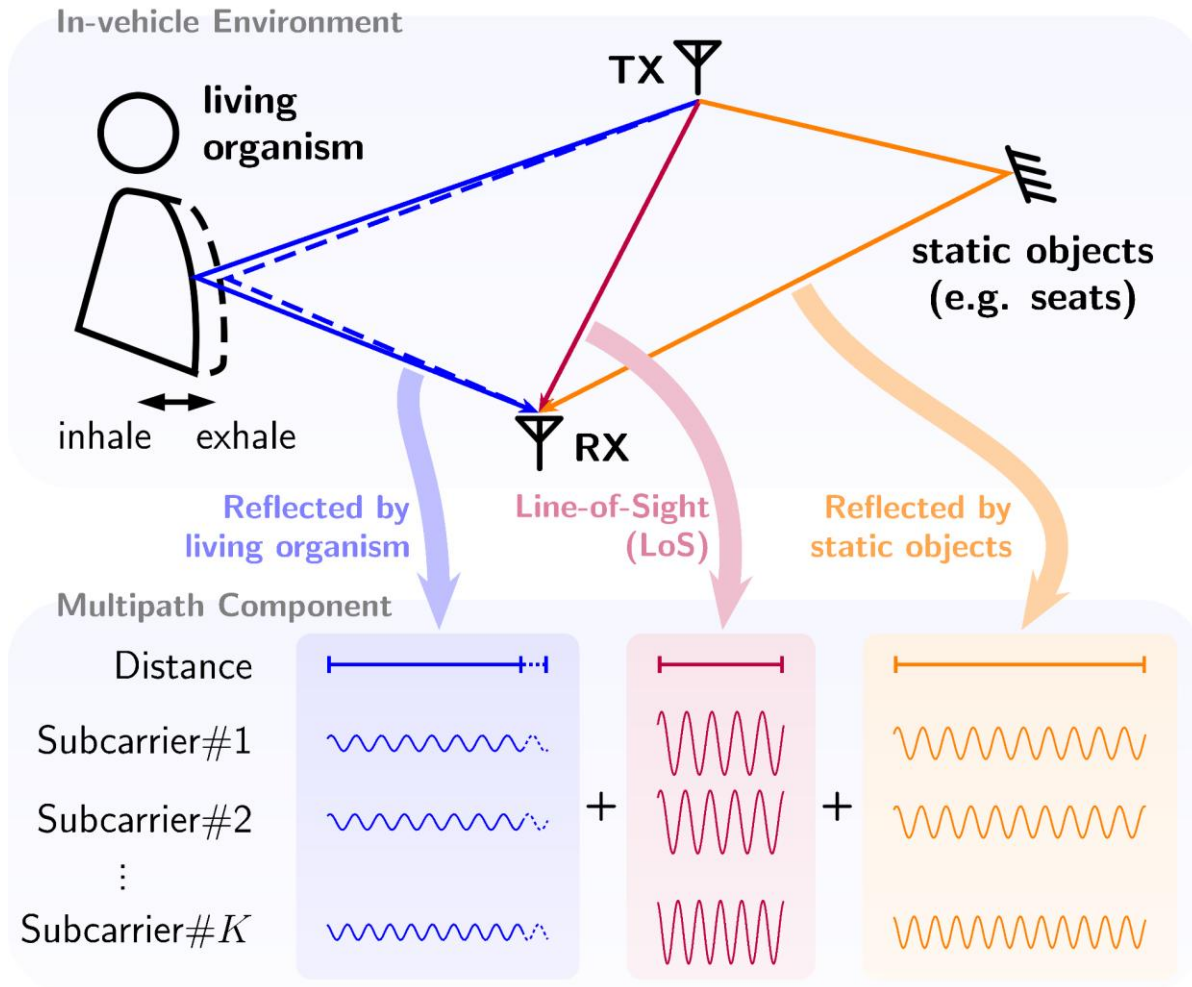


Typical multipath environment

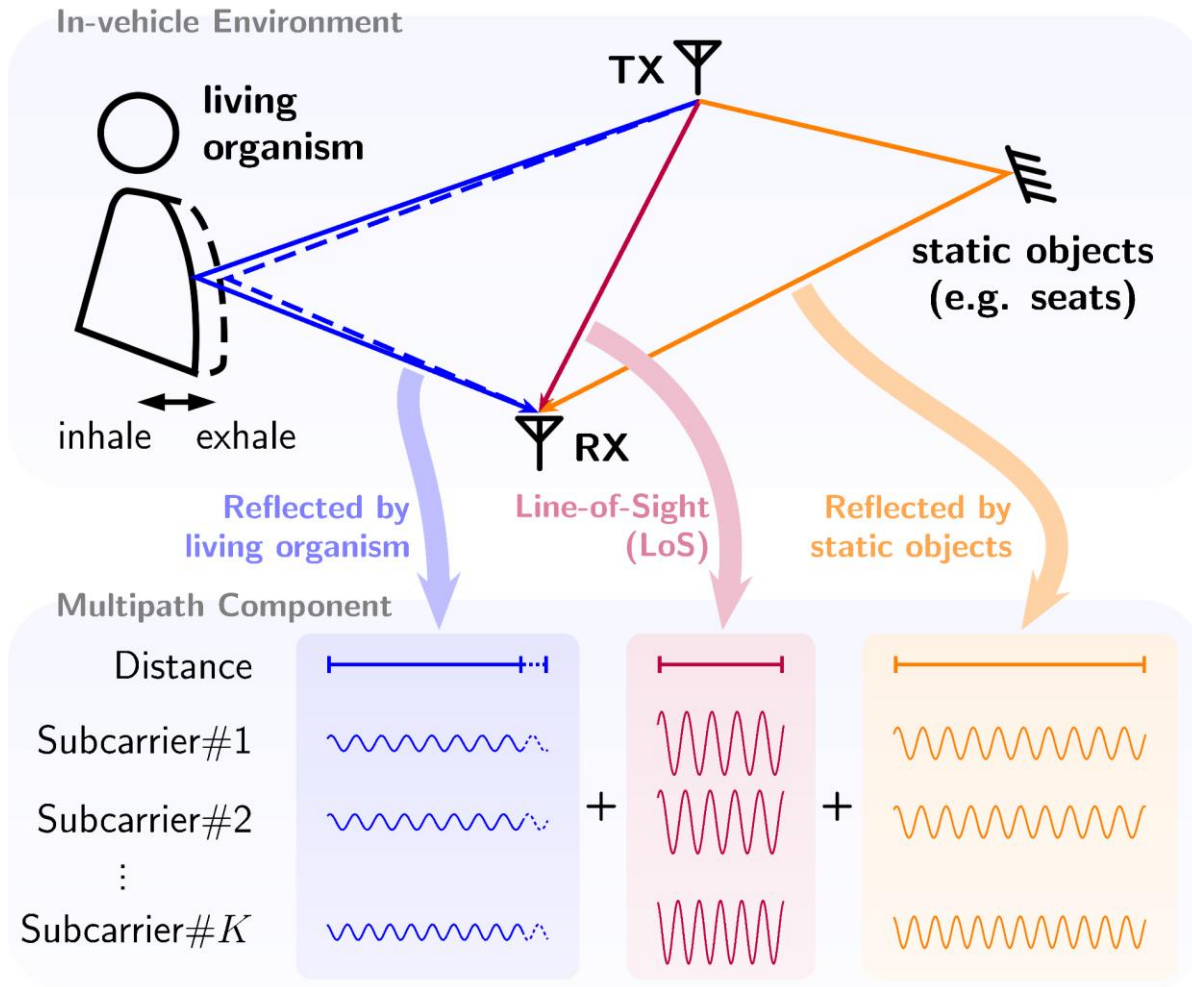
$$H(t) = \sum_{l=1}^L H_l(t)$$

obtained the CSI component for single path

CSI of complex multipath conditions



Model for Time-varying Envs in Subcarriers



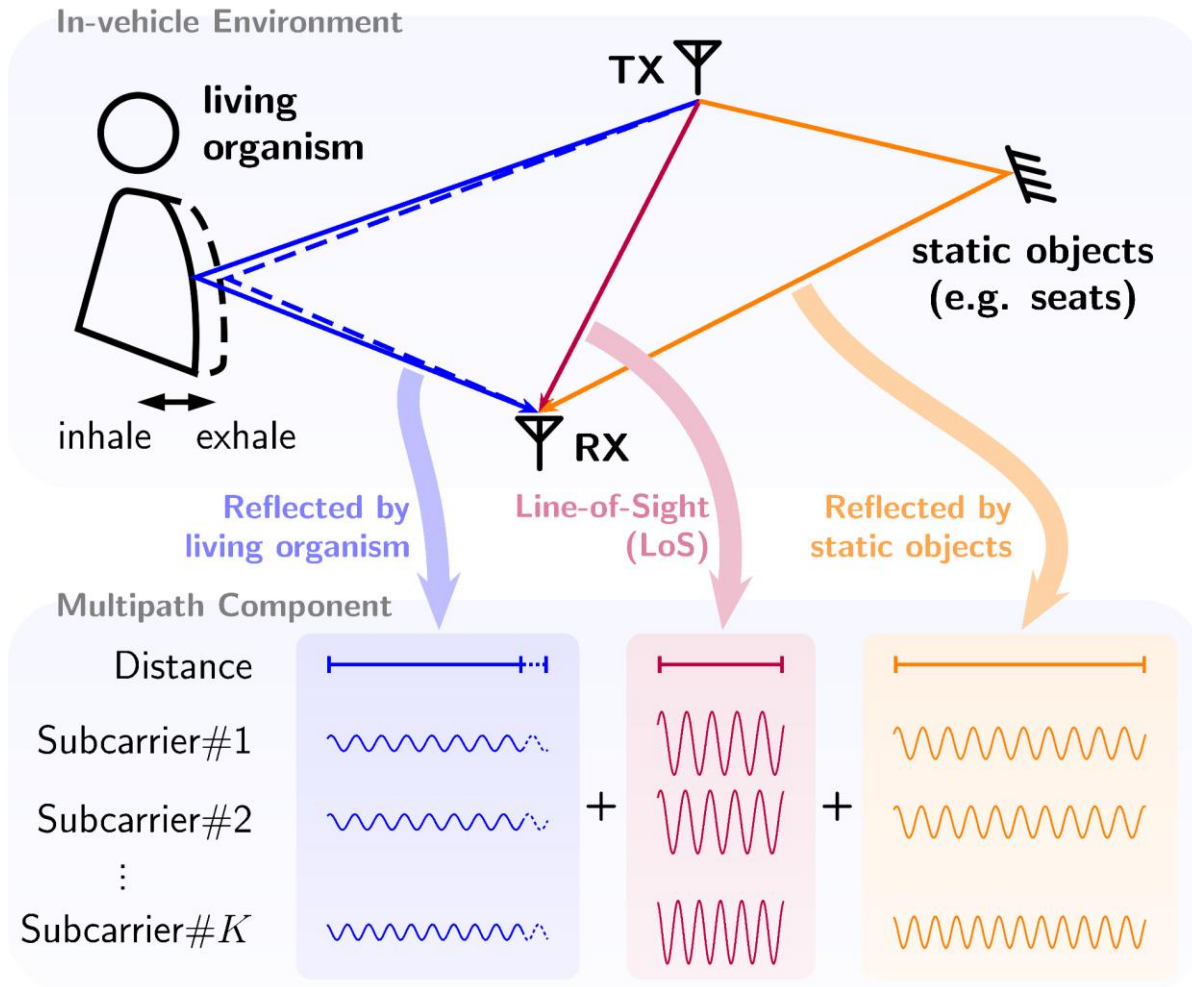
$$\begin{aligned}
 H(t) &= \sum_{l=1}^L H_l(t) \\
 &= G_{Tx}(t)G_{Rx}(t) \sum_{l=1}^L \left\{ \prod_{m=1}^M \frac{\sigma_{l,m}}{4\pi R_{l,m}^2(t)} \exp \left[j \left(\frac{2\pi f_i}{c} \sum_{m=1}^M R_{l,m}(t) + (M-1)\pi \right) \right] \right\}
 \end{aligned}$$

near-stable

stable

time-varying

Model for Time-varying Envs in Subcarriers



$$H(t) = \sum_{l=1}^L H_l(t)$$

$$= G_{Tx}(t)G_{Rx}(t) \sum_{l=1}^L \left\{ \prod_{m=1}^M \frac{\sigma_{l,m}}{4\pi R_{l,m}^2(t)} \right.$$

$$\left. \exp \left[j \left(\frac{2\pi f_i}{c} \sum_{m=1}^M R_{l,m}(t) + (M-1)\pi \right) \right] \right\}$$

extract the distance variation

$$\Delta R_l(t) = \sum_{m=1}^M R_{l,m}(t) - R_{l,m}(t_0)$$

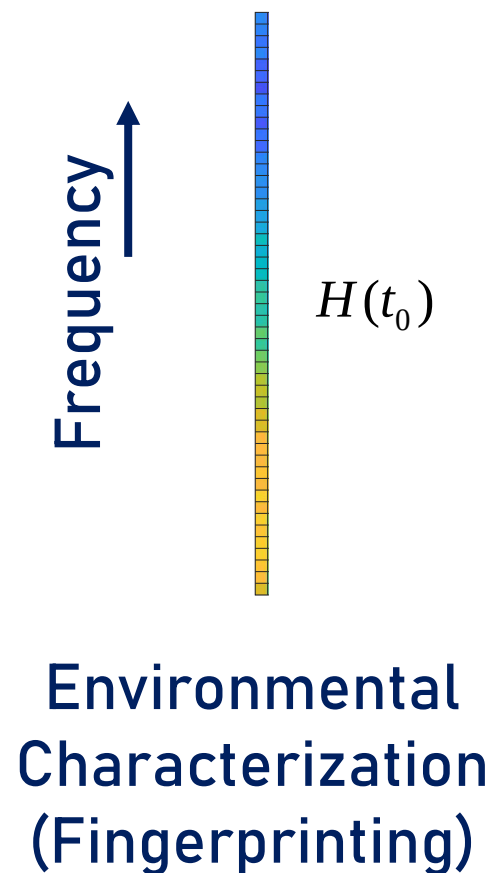
$$H(t) = G_{Tx}(t)G_{Rx}(t) \sum_{l=1}^L H_l \exp \left[j \frac{2\pi f_i}{c} \Delta R_l(t) \right]$$

Factors affecting the variation of CSI

Model for Time-varying Envs in Subcarriers



$$H(t) = G_{Tx}(t)G_{Rx}(t)\sum_{l=1}^L H_l \exp\left[j\frac{2\pi f_i}{c}\Delta R_l(t)\right]$$



the relationship between fingerprints and the environment

a given
multipath
environment



an unique
fingerprint



time-varying CSI
(a series of
fingerprints)

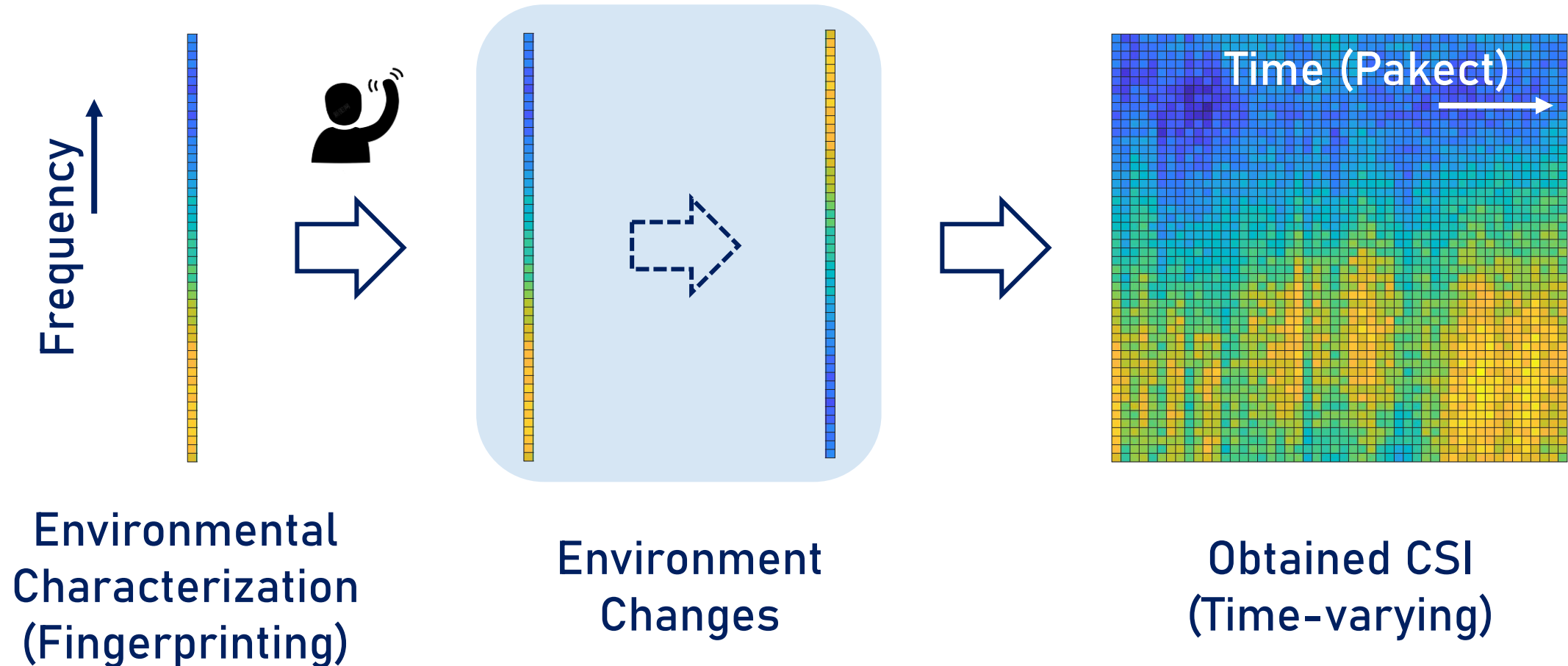


changes in
multipath
environment

Model for Time-varying Envs in Subcarriers



$$H(t) = G_{Tx}(t)G_{Rx}(t)\sum_{l=1}^L H_l \exp\left[j\frac{2\pi f_i}{c}\Delta R_l(t)\right]$$

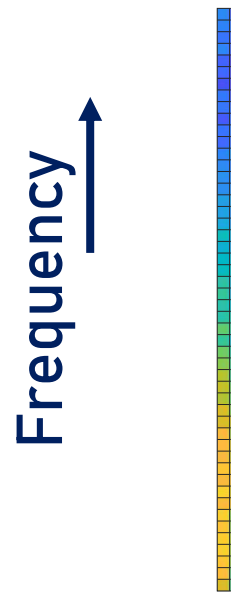


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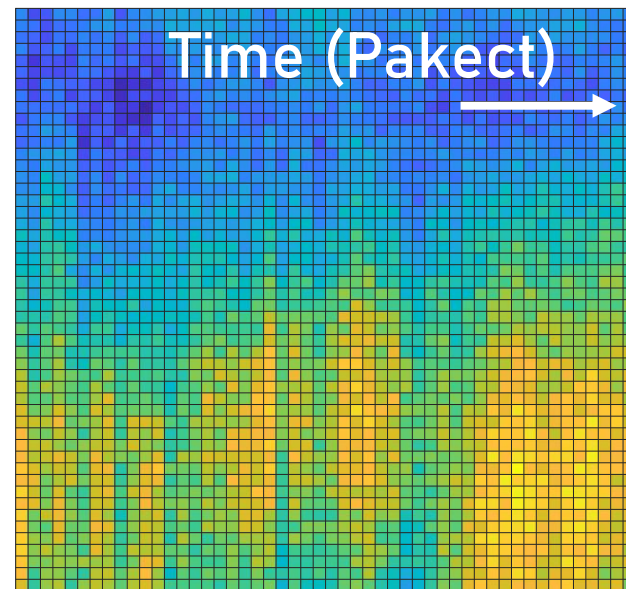


$$H(t) = G_{Tx}(t)G_{Rx}(t)\sum_{l=1}^L H_l \exp\left[j\frac{2\pi f_i}{c}\Delta R_l(t)\right]$$

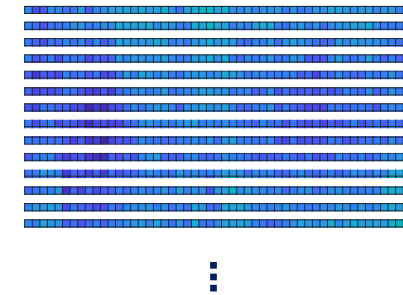
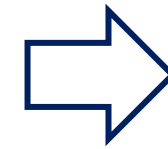
$\left[j\frac{2\pi f_i}{c}\Delta R_l(t) \right]$ treated as a constant
 $\Delta R_l(t)$ treated as a variable



Environmental
Characterization
(Fingerprinting)



Obtained CSI
(Time-varying)



Time Dimension
(with Long Time Window)

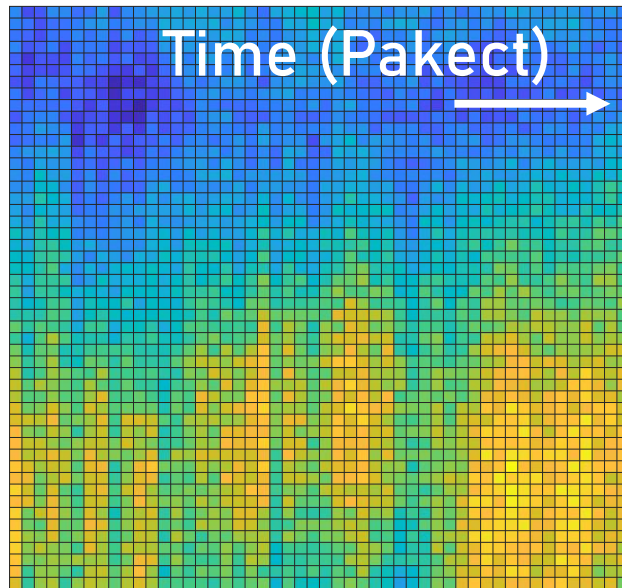
- Subcarrier is treated independently
- It takes a long time sequence
to observe changes in the signal
when the movement is small

Model for Time-varying Envs in Subcarriers



$$H(t) = G_{Tx}(t)G_{Rx}(t)\sum_{l=1}^L H_l \exp\left[j\frac{2\pi f_i}{c}\Delta R_l(t)\right]$$

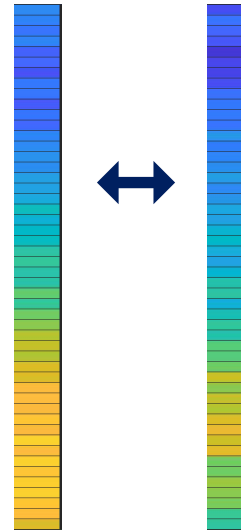
treated as a variable
treated as a constant



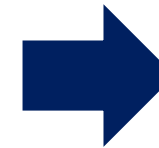
Obtained CSI
(Time-varying)



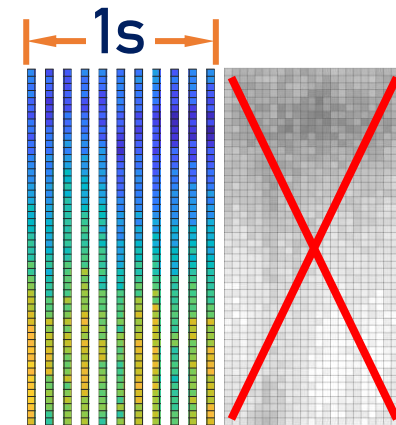
small
environmental
changes



Continuous variations in
subcarrier dimension

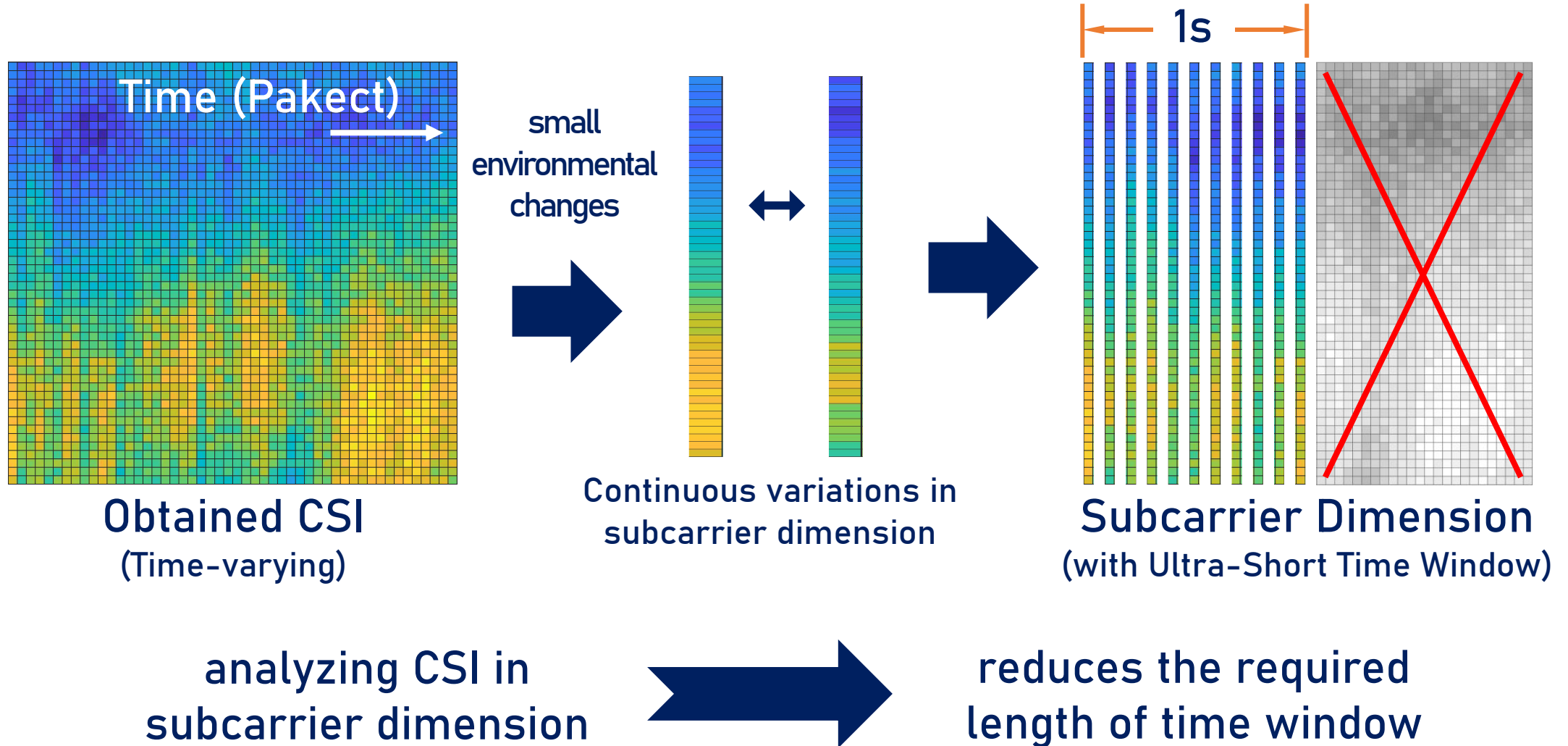


Time Dimension
(with Long Time Window)



Subcarrier Dimension
(with Ultra-Short Time Window)

Model for Time-varying Envs in Subcarriers



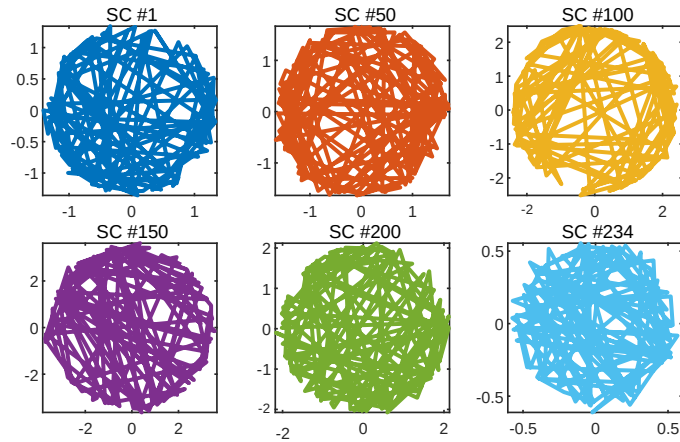
RapidPD Design



RapidPD Design



Raw CSI with Phase Offset



presents irregularities in the complex plane

CSI Pre-
processing

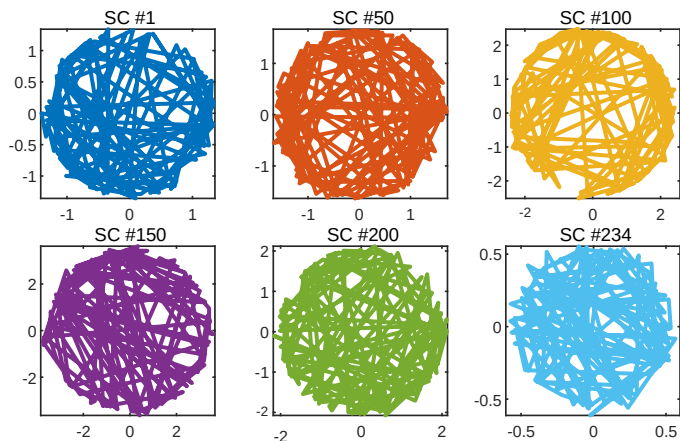
Motion
Target
Detector

Presence
Detection
Indicator

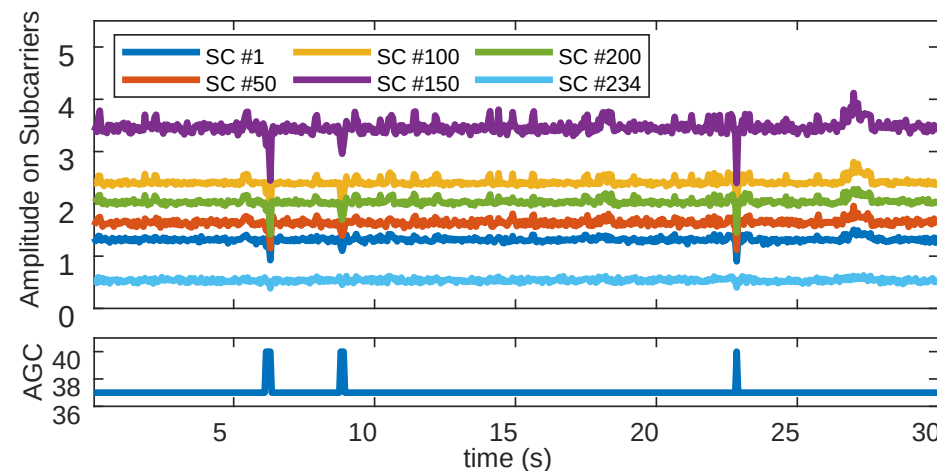
RapidPD Design



Raw CSI with Phase Offset



Unrepaired CSI with AGC



Analyze only the magnitude of the CSI

- maintain real-time performance
- minimize computational costs

CSI Pre-
processing

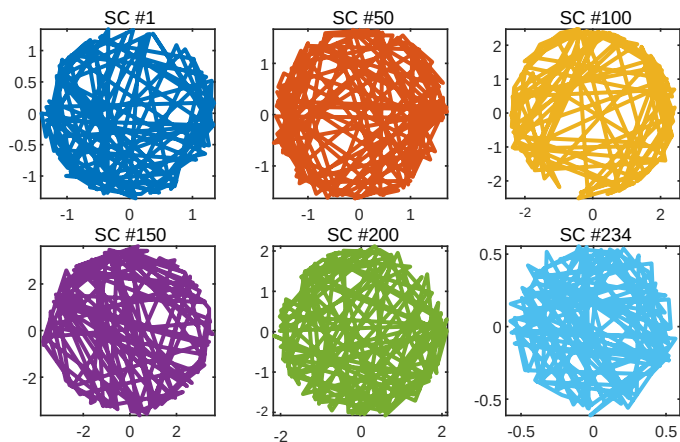
Motion
Target
Detector

Presence
Detection
Indicator

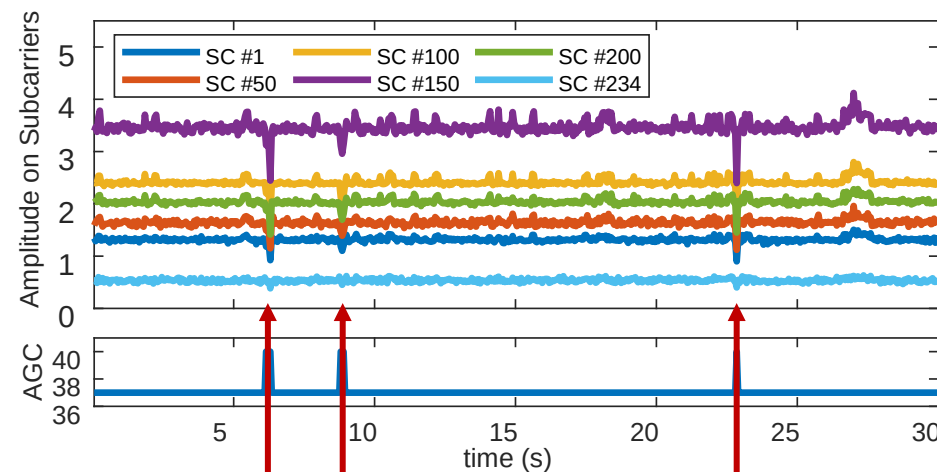
RapidPD Design



Raw CSI with Phase Offset



Unrepaired CSI with AGC



exhibits several spikes

CSI Pre-
processing

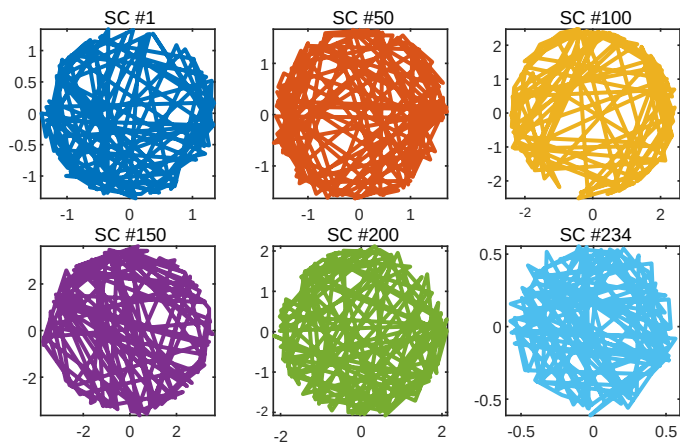
Motion
Target
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Indicator

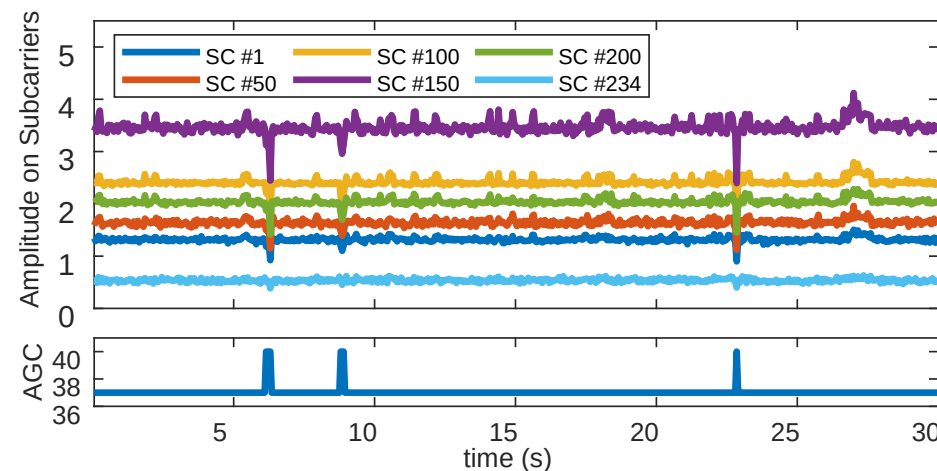
RapidPD Design



Raw CSI with Phase Offset

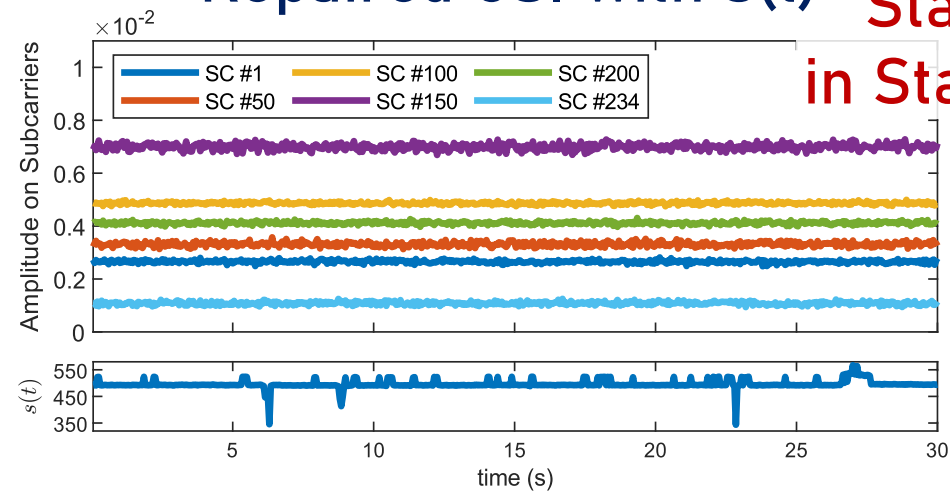


Unrepaired CSI with AGC



Repaired CSI with $s(t)$

Stability
in Static Env



CSI Pre-
processing

Motion
Target
Detector

Presence
Detection
Indicator

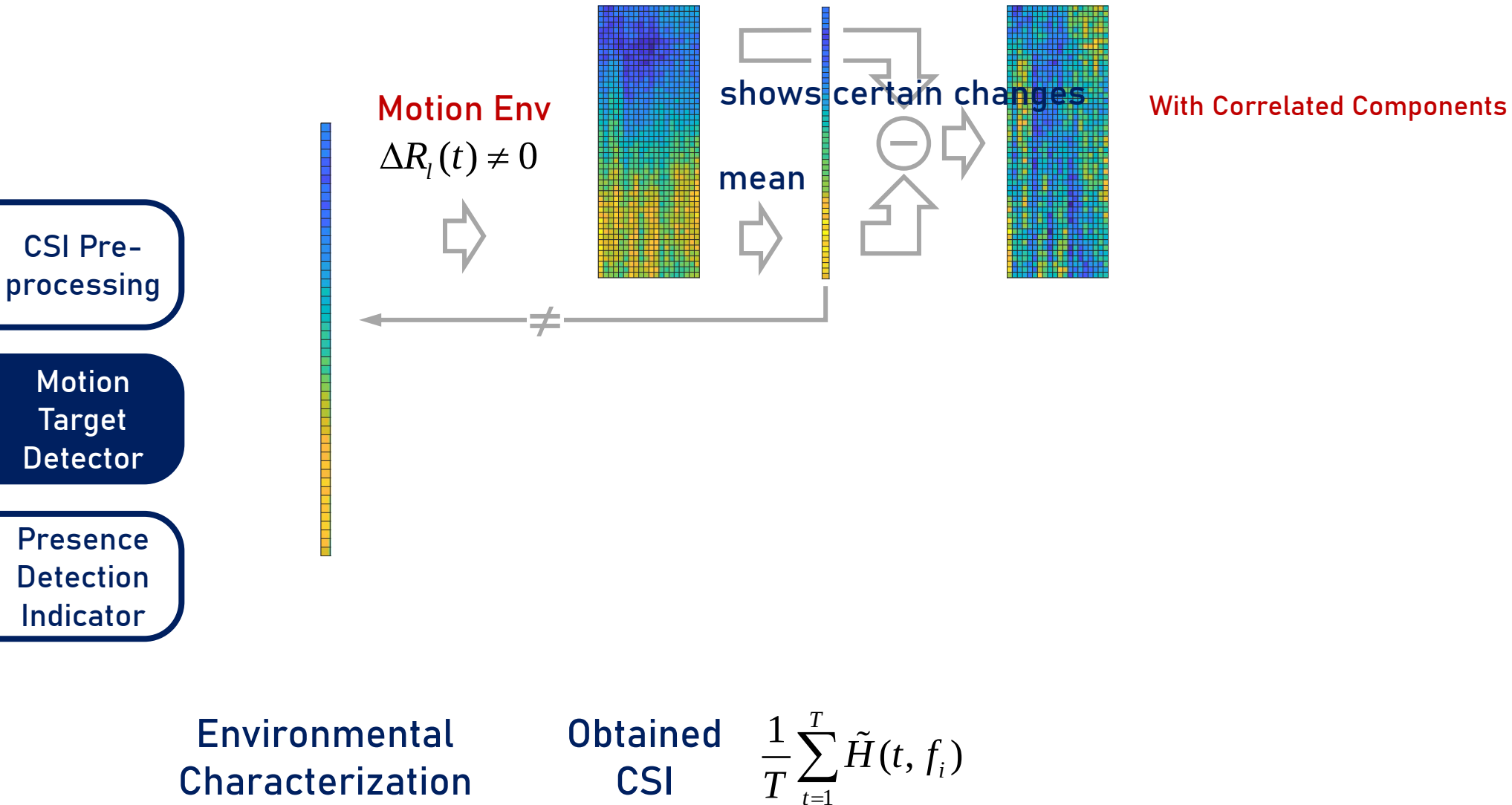
$$s(t) = \sum_{i \in \Omega_F} |H(t, f_i)|$$

$$\tilde{H}(t, f_i) = \frac{|H(t, f_i)|}{s(t)}$$

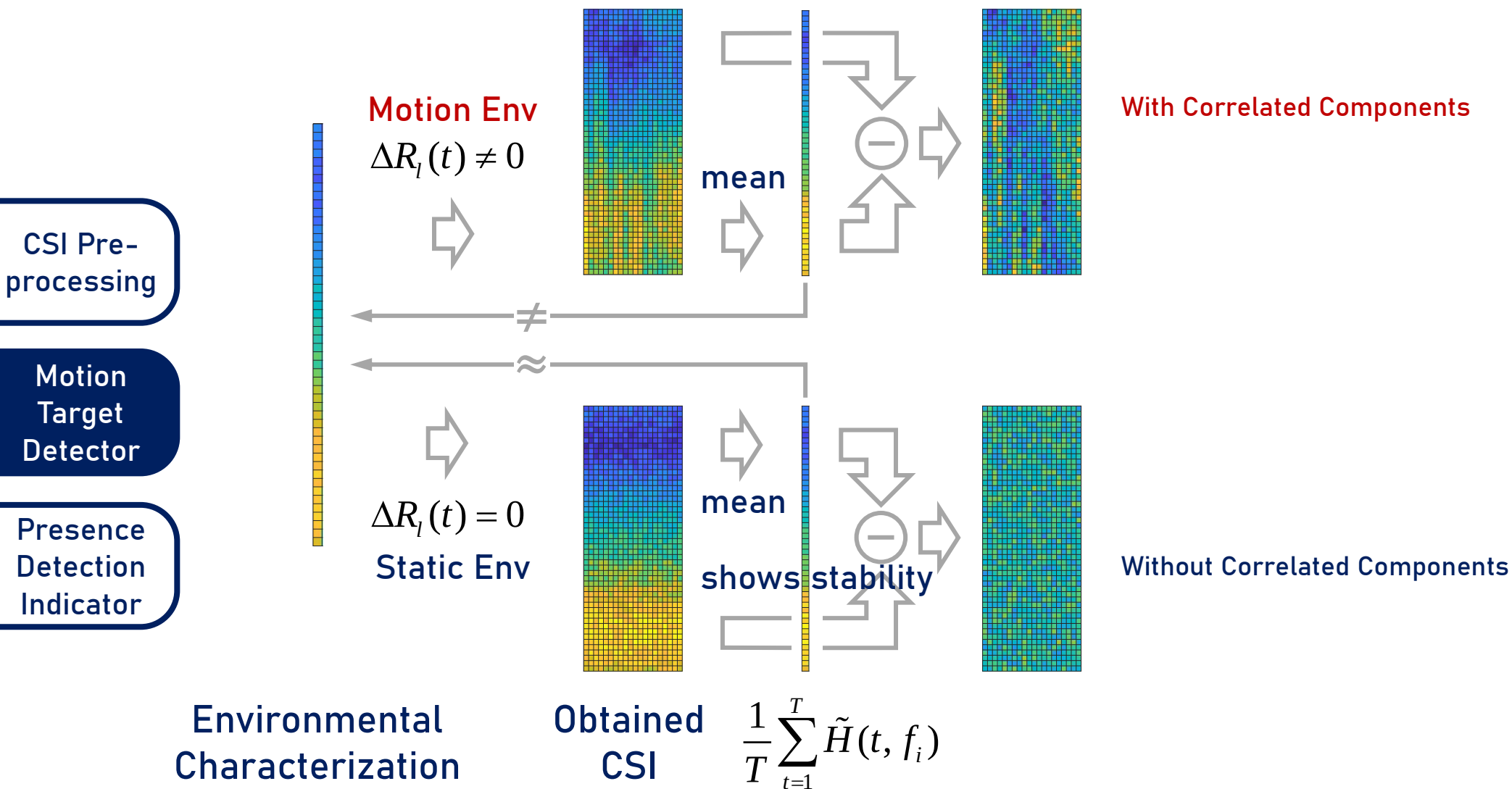
$$= \frac{G(t)}{s(t)} \left| \sum_{l=1}^L H'_l(f_i) e^{j \frac{2\pi f_i}{c} \Delta R_l(t)} \right|$$

normalize CSI magnitude

RapidPD Design



RapidPD Design



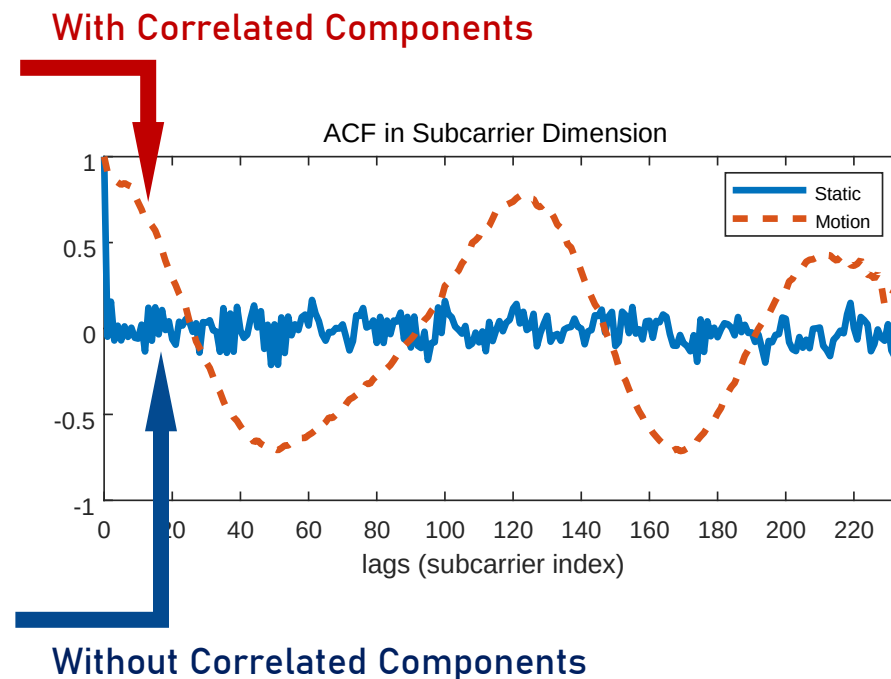
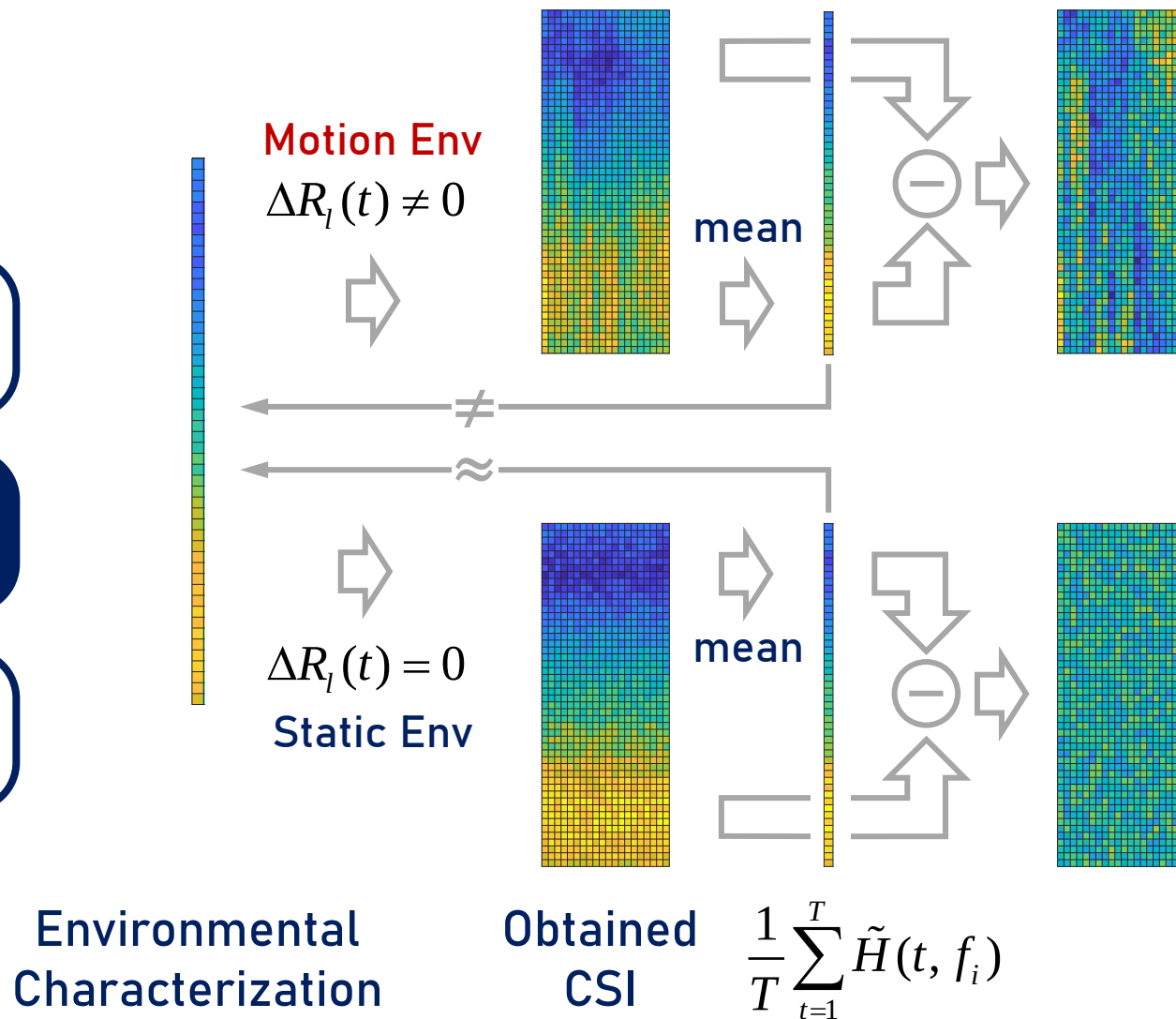
RapidPD Design



CSI Pre-processing

Motion Target Detector

Presence Detection Indicator



Autocorrelation Function in Subcarrier Dimension

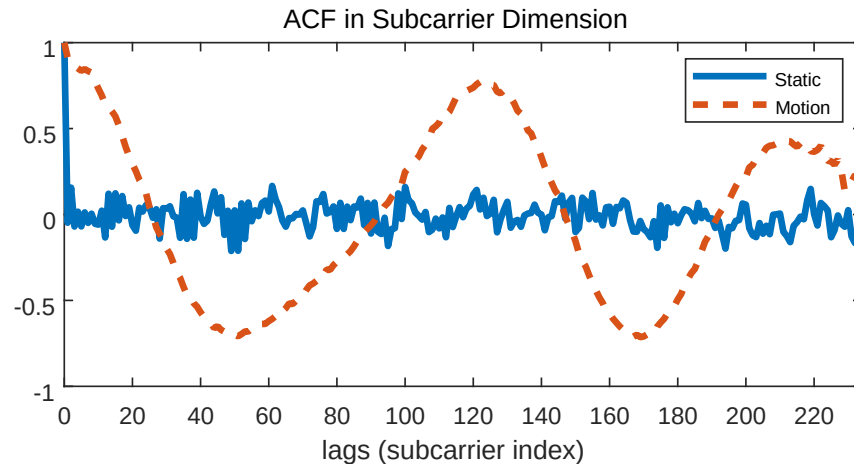
RapidPD Design



CSI Pre-
processing

Motion
Target
Detector

Presence
Detection
Indicator



The method proposed above

Strong Signal



Capture the
environment
changes

Weak Signal



Become
ineffective

RapidPD Design

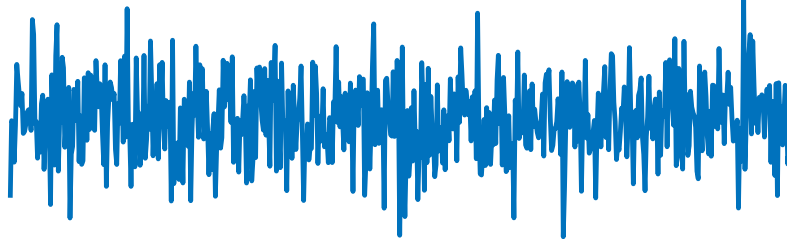


CSI Pre-
processing

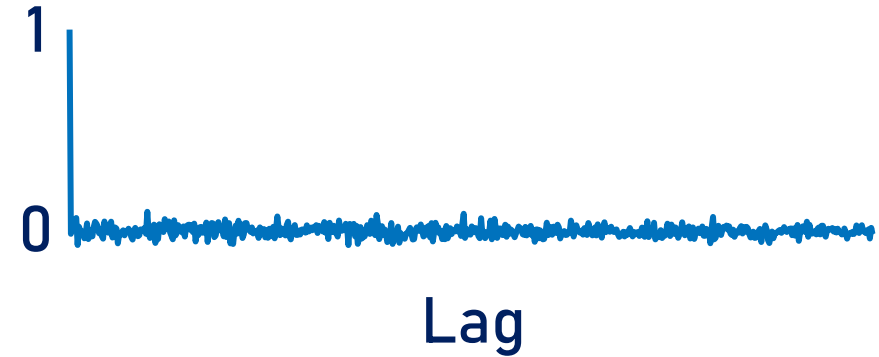
Motion
Target
Detector

Presence
Detection
Indicator

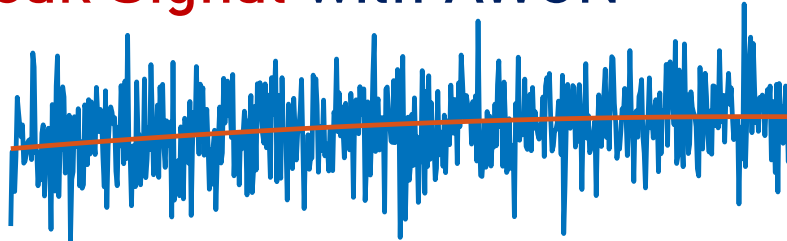
Additive Gaussian White Noise



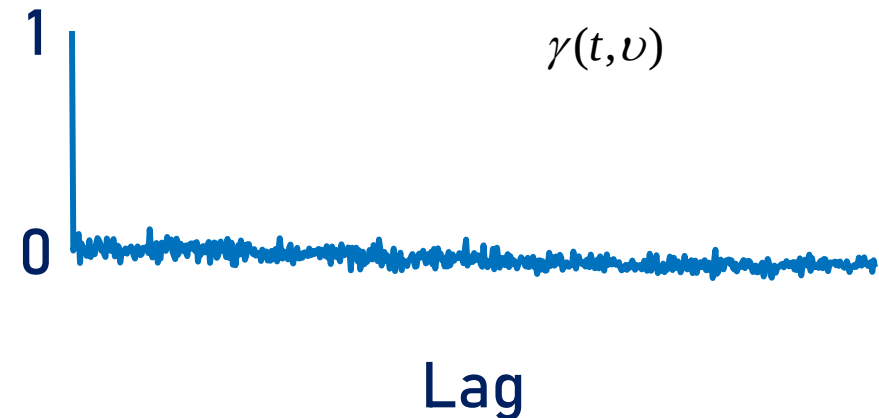
ACF



Weak Signal with AWGN



ACF



RapidPD Design

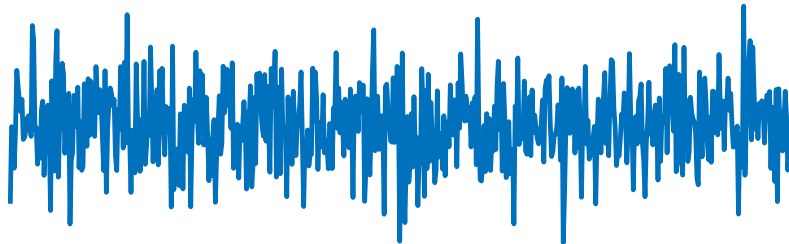


CSI Pre-
processing

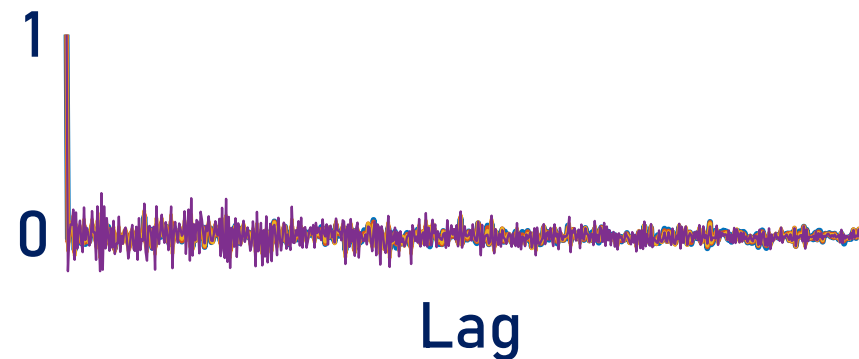
Motion
Target
Detector

Presence
Detection
Indicator

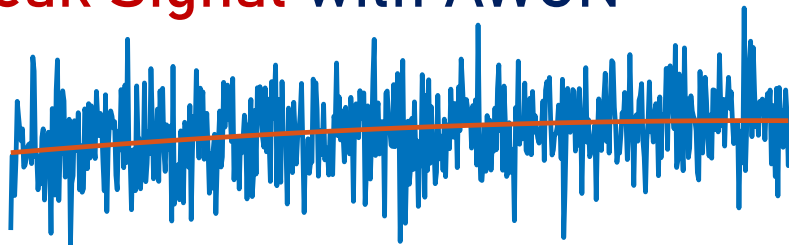
Additive Gaussian White Noise



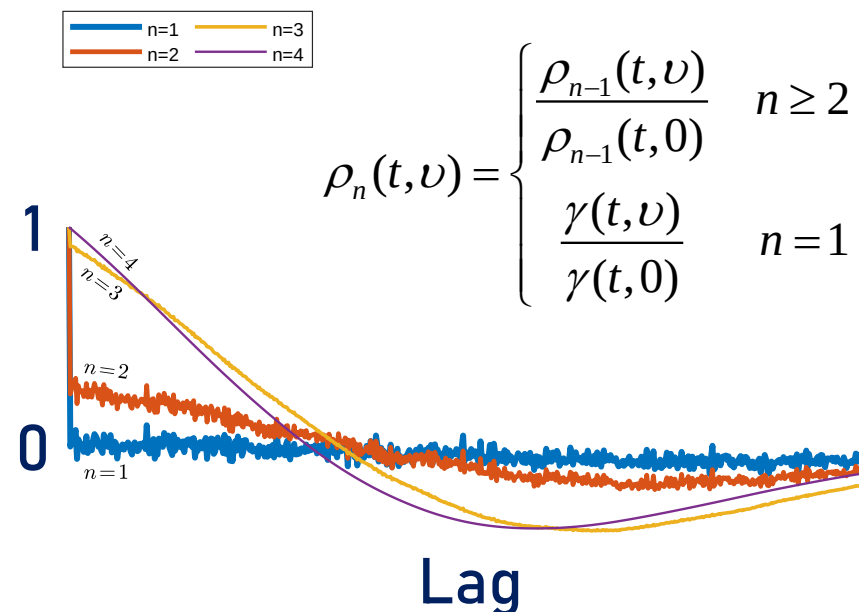
Multi-
ACF



Weak Signal with AWGN



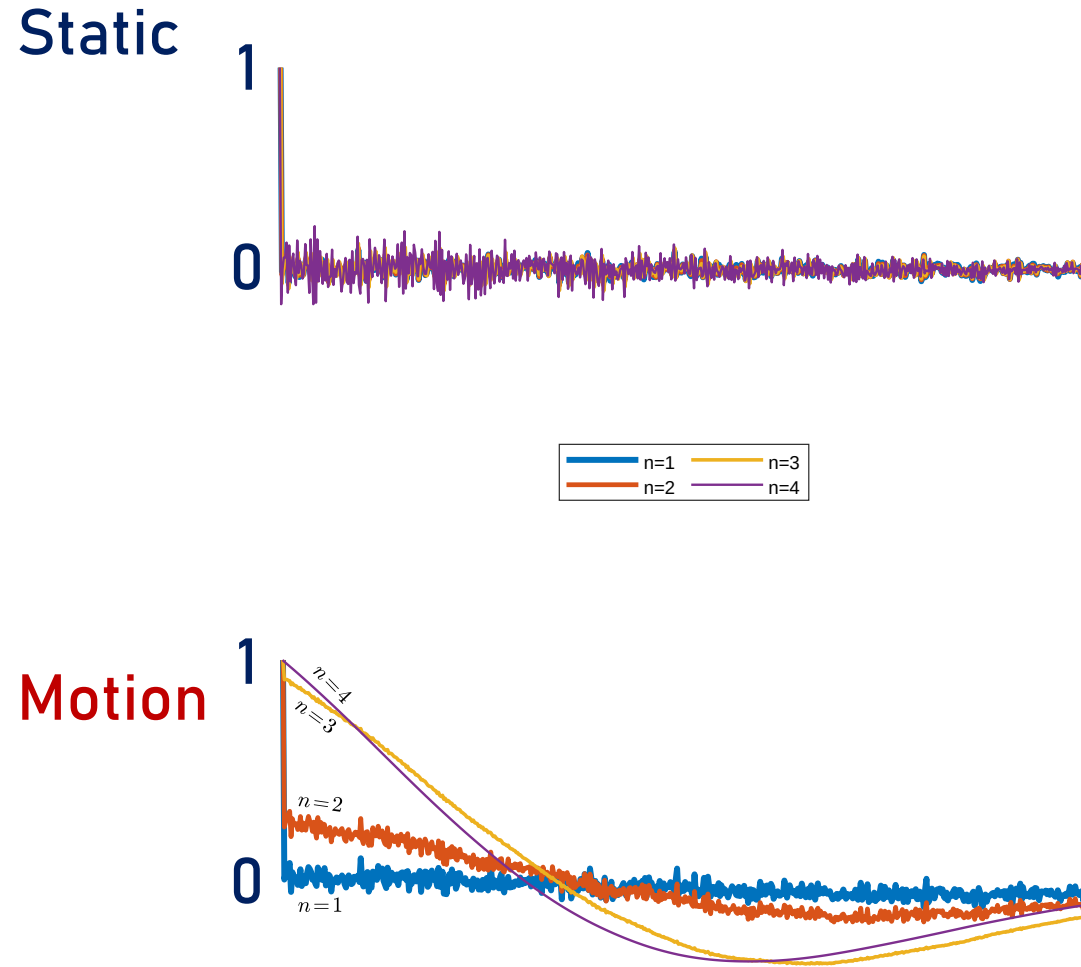
Multi-
ACF



CSI Pre-
processing

Motion
Target
Detector

Presence
Detection
Indicator



Multi-layer Autocorrelation

- mathematical expression:

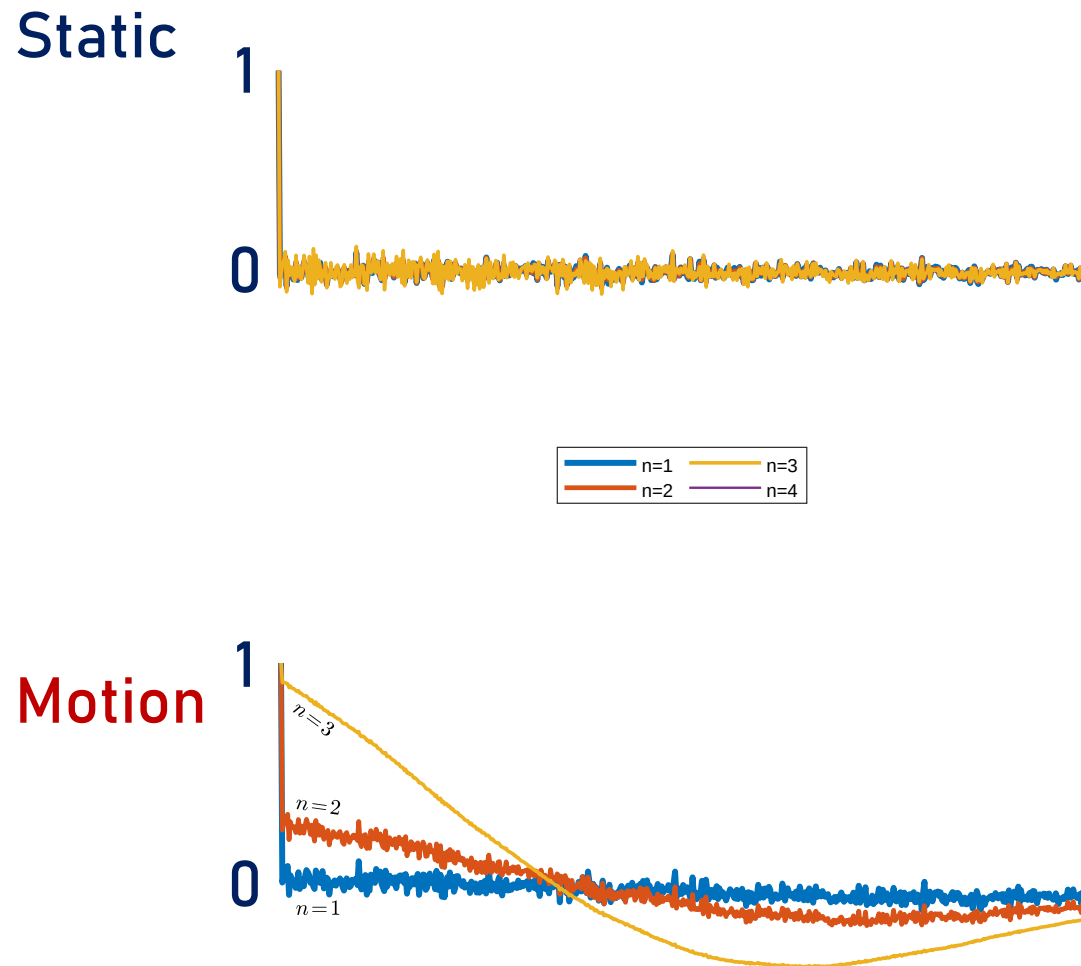
$$\rho_n(t, \nu) = \begin{cases} \frac{\rho_{n-1}(t, \nu)}{\rho_{n-1}(t, 0)} & n \geq 2 \\ \frac{\gamma(t, \nu)}{\gamma(t, 0)} & n = 1 \end{cases}$$

↑ represents
sample auto-covariance

CSI Pre-
processing

Motion
Target
Detector

Presence
Detection
Indicator



Multi-layer Autocorrelation

- mathematical expression:

$$\rho_n(t, \nu) = \begin{cases} \frac{\rho_{n-1}(t, \nu)}{\rho_{n-1}(t, 0)} & n \geq 2 \\ \frac{\gamma(t, \nu)}{\gamma(t, 0)} & n = 1 \end{cases}$$

↑ represents
sample auto-covariance

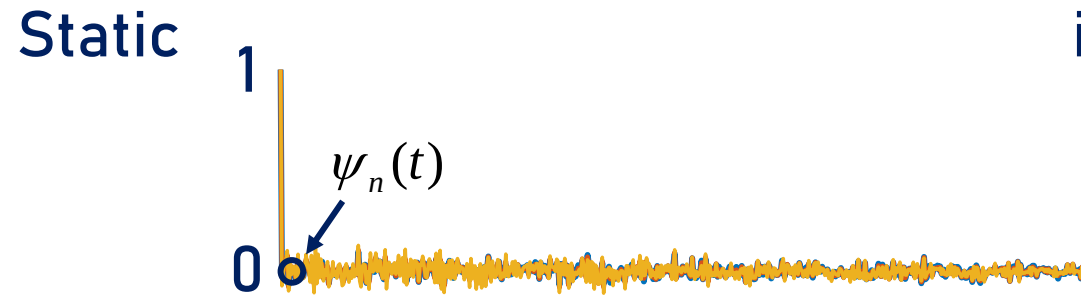
- default value:

$$n = 3$$

CSI Pre-
processing

Motion
Target
Detector

Presence
Detection
Indicator

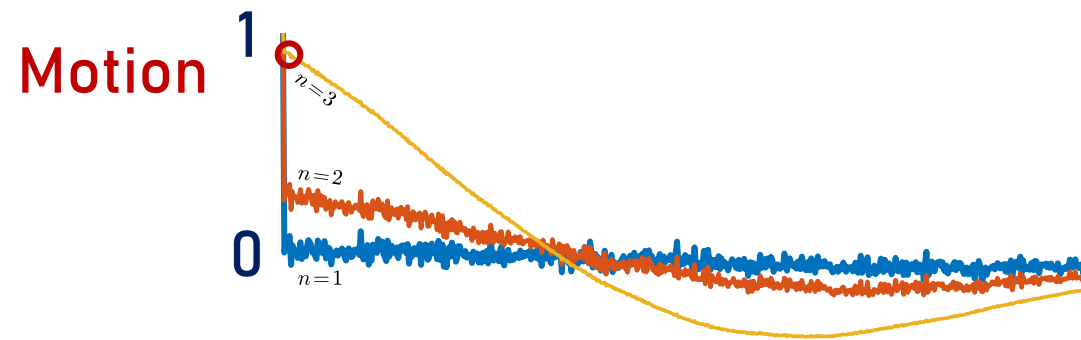


indicator based on multi-layer ACF

- A CFR sample in a data stream

$$\psi_n(t) = \rho_n(t, \Delta f)$$

↑
subcarrier spacing



RapidPD Design

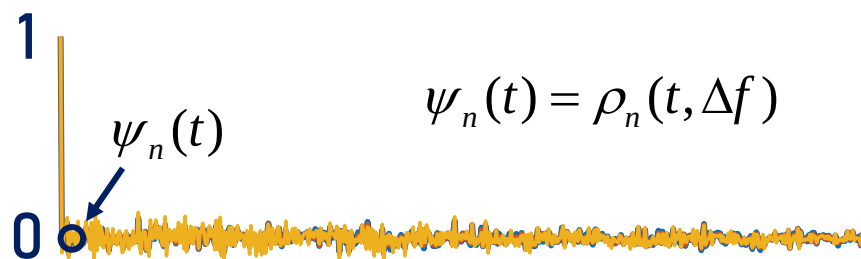


CSI Pre-processing

Motion Target Detector

Presence Detection Indicator

Static



Sum Packets

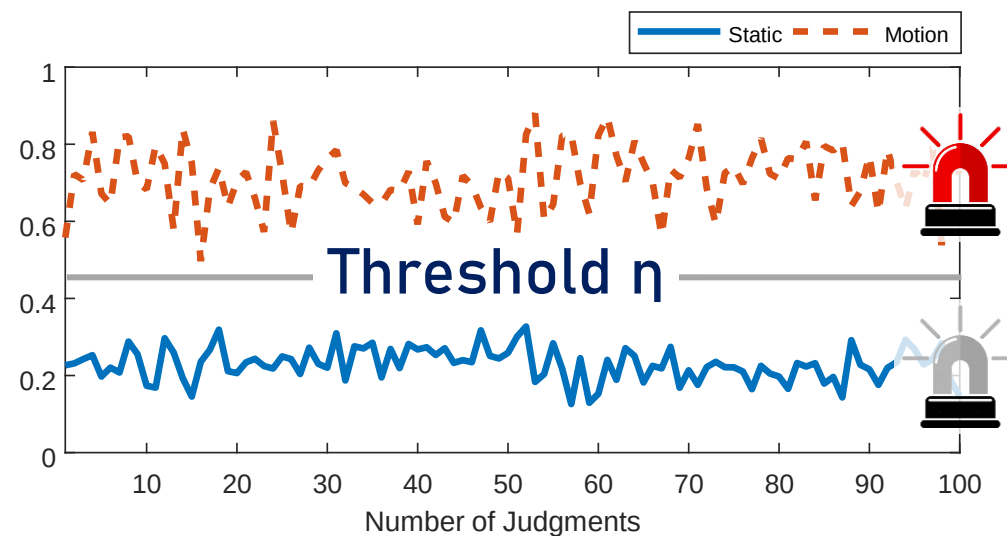
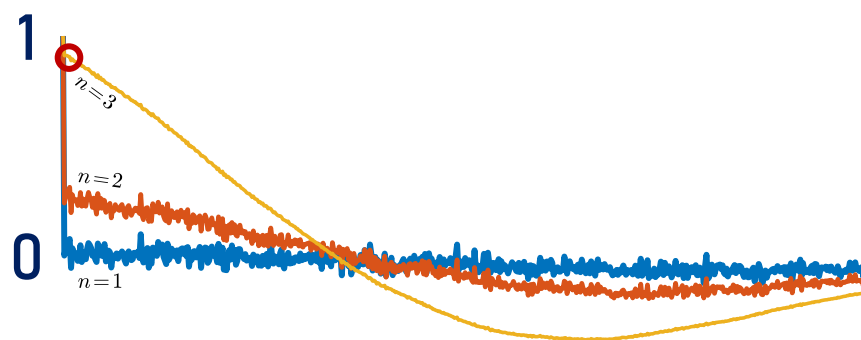
$$\phi = \sum_{t=1}^T \psi_n(t)$$



Sum Streams

$$\Phi = \sum \phi$$

Motion



Motion Statistics
in Subcarrier Dimension

Implementation



Hardware platform

- 2Tx 1Rx
- 20 Hz sampling rate
- 155 channel (center freq of 5775 MHz)
- 80 MHz bandwidth



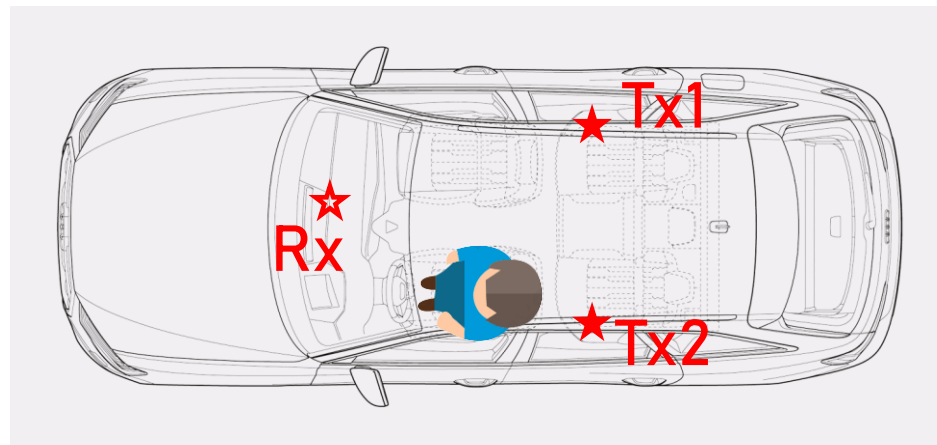
Position of the antenna

- 2Tx are deployed above the rear doors
- Rx is placed near the center console
- All antennas are PCB omnidirectional antennas.

Implementation



Positions for Living Beings



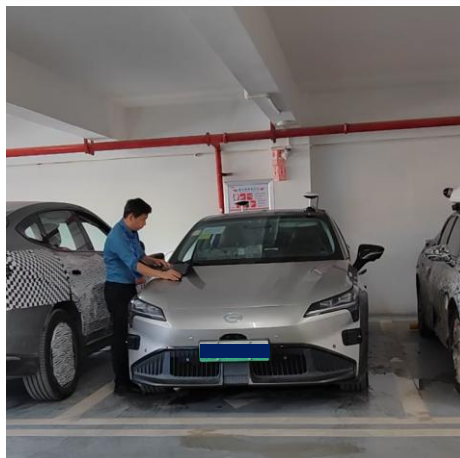
Schematic of the Experiment

- 4 Living Being Cases
 - 1) empty
 - 2) human presence(include child and infant)
 - 3) dog presence
 - 4) cat presence
- 11 Living Being Positions
 - 1)-5) sit on seat
 - 6)-10) located in the footwell
 - 11) lying across the rear seat

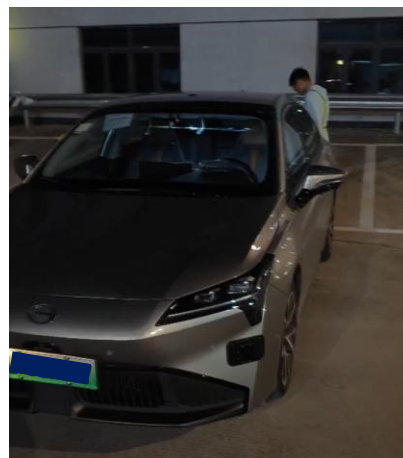
Experiment



Environment



parking structures



outdoor open spaces



roadsides

.....

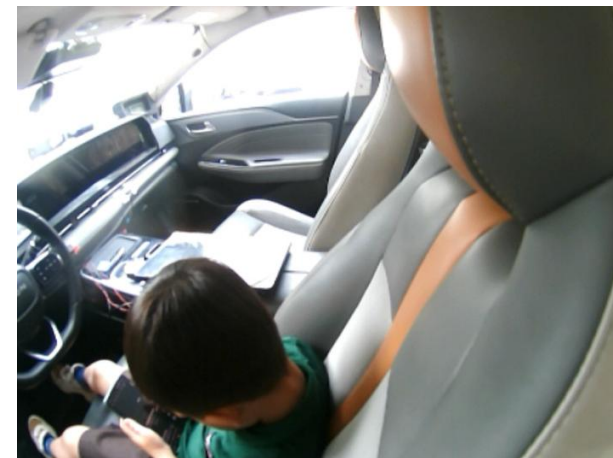
Living Beings



pet cat



pet dog



child

.....

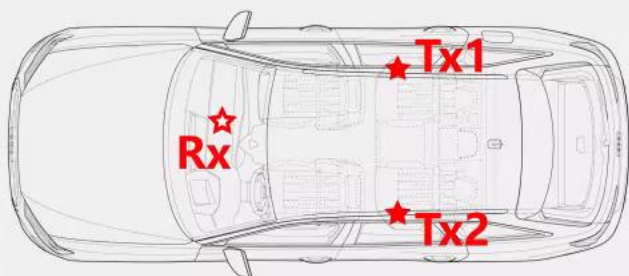
Experiment



Real-time Video **Empty**



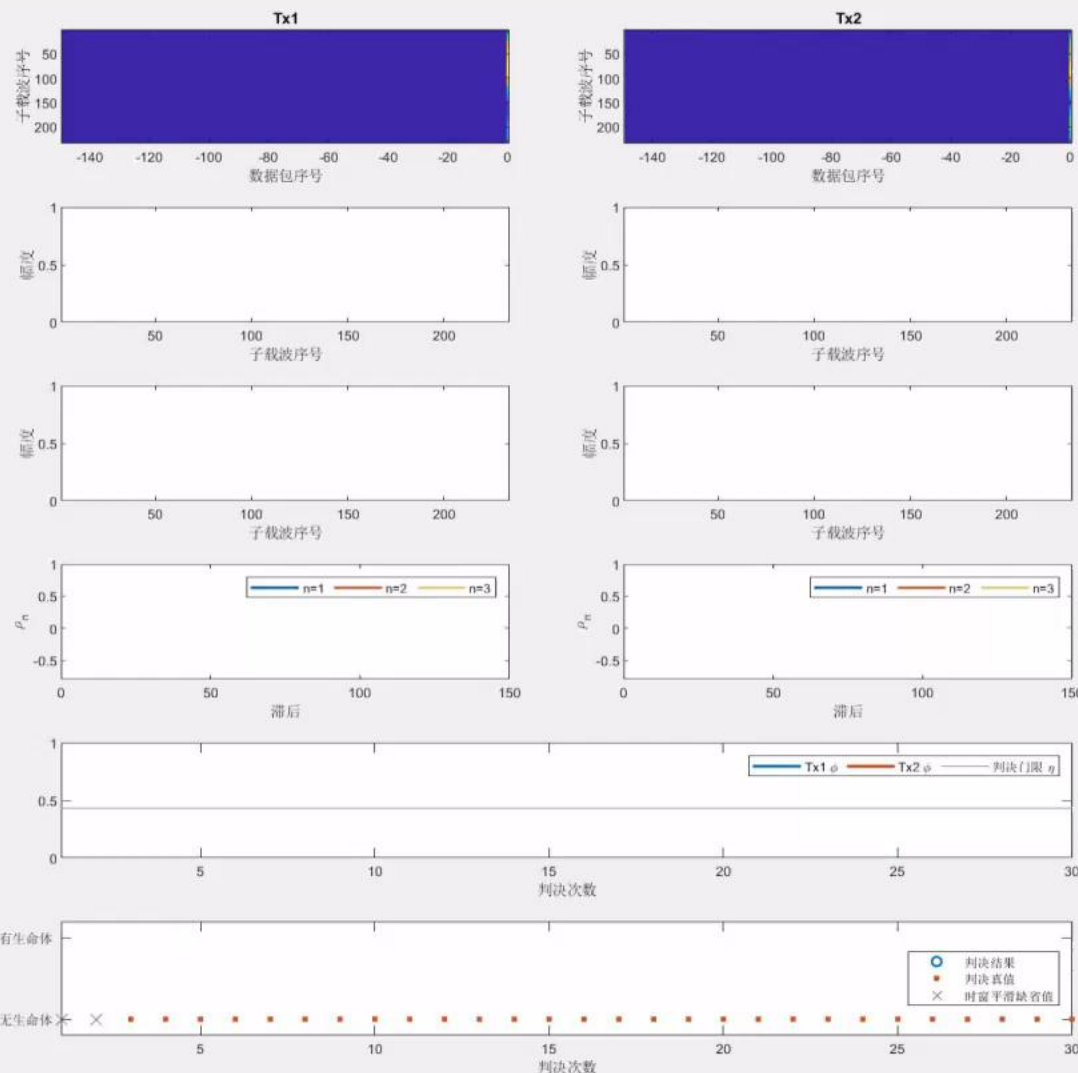
Scenario Setup Diagram



Alert

Living
Beings
Exist

User Interface



Original
Data

Normalized
Data

Multi-layer
ACF

Motion
Statistics

Judgment

Overall Performance



Evaluation results in real-world tests

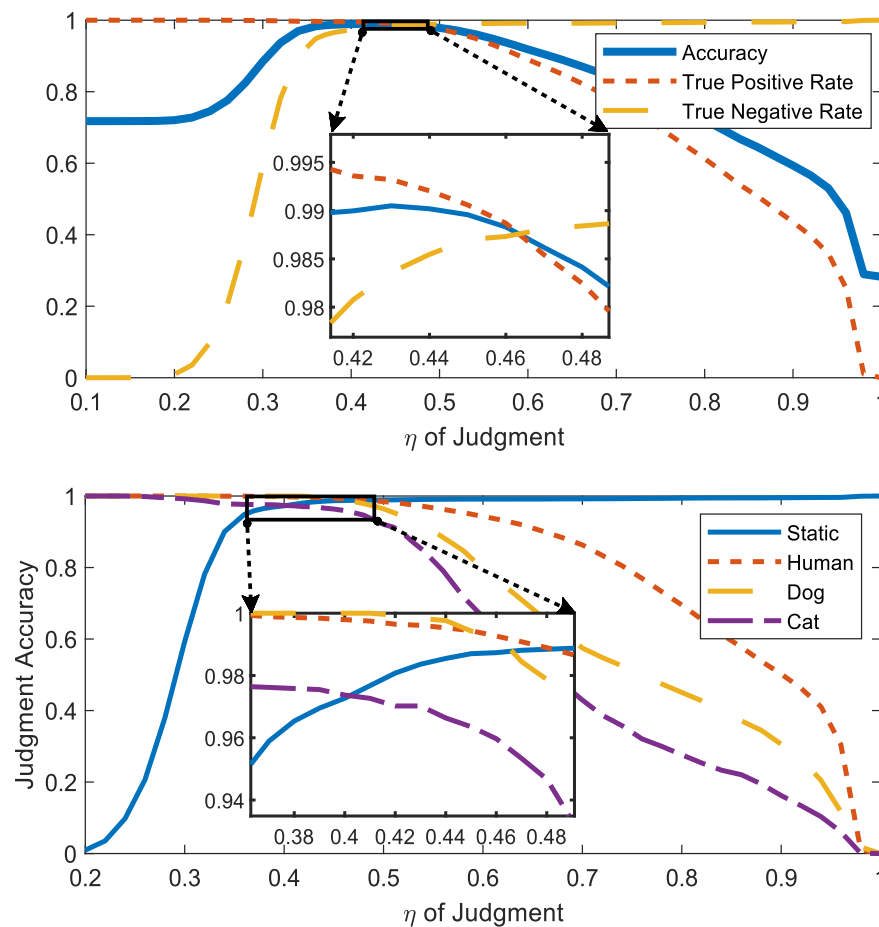
Actual scenarios	Static	0.9836	0.0164
	Human	0.0039	0.9961
	Dog	0.0017	0.9983
	Cat	0.0298	0.9702
Predicted scenarios		Static	Living Organism

- High detection accuracy in 4 cases
- Overall detection accuracy reaches **99.05%**
- True positive rate (TPR) reaches **99.32%**
- The false positive rate is as low as **1.64%**

Overall Performance



Relationship between performance and threshold η

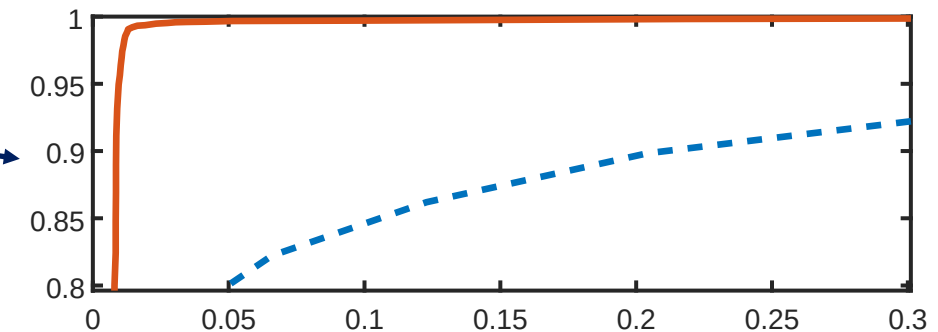
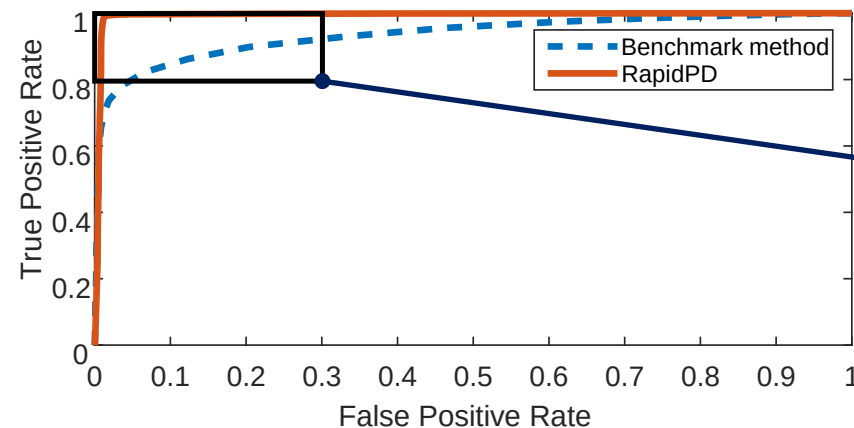


- A trade-off exists in system performance.
- threshold of $\eta=0.43$ is selected to ensure a high recall while maintaining a low false alarm
- the detection performance for human presence remains above **95%** over a wide range of threshold (about $\eta < 0.6$).

Comparison with Existing Works



(Method in 1-second window)	TPR	FPR
Benchmark Method (time-domain)	89.83%	20.41%
RapidPD without Multi-layer ACF	99.01%	3.16%
RapidPD (with Multi-layer ACF)	99.32%	1.64%



Significantly superior performance is expected to facilitate the deployment of CPD.

RapidPD

Rapid Human and Pet Presence Detection System for Smart Vehicles via Wi-Fi

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